Chapter 1. The number 1

This chapter is about the number 1. In Roman numerals it is written I, in binary it is 1, and in base eight it is 1. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

1 is neither prime nor composite. It has exactly one divisor, itself. Every whole number greater than 1 is either prime or can be written as a product of primes in exactly one way, apart from the order of the factors.

The sum of the divisors of 1 that are smaller than 1 is 0. Because this sum is smaller than 1, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 1 is 1 and its cube is 1. 1 is itself a perfect square, because 1 × 1 = 1. The sum of its digits is 1, so 1 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 1 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 1: find two whole numbers whose product is 2, then two whole numbers whose sum is 8. Then write 1 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 1 is odd. A hint: divisors usually come in pairs whose product is 1, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 1 that has the same number of divisors, or test whether 2 and 2 share any common factor with 1. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 2. The number 2

This chapter is about the number 2. In Roman numerals it is written II, in binary it is 10, and in base eight it is 2. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

2 is a prime number. Its only divisors are 1 and 2. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 2 counters in equal rows, the only way is a single row of 2. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 2 is one of them.

The sum of the divisors of 2 that are smaller than 2 is 1. Because this sum is smaller than 2, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 2 is 4 and its cube is 8. 2 is not a perfect square: it lies between 1 and 4. The sum of its digits is 2, so 2 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 2 is even, so it can be shared equally between two people.

Exercise for chapter 2: find two whole numbers whose product is 4, then two whole numbers whose sum is 9. Then write 2 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 2 is even. A hint: divisors usually come in pairs whose product is 2, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 2 that has the same number of divisors, or test whether 3 and 1 share any common factor with 2. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 3. The number 3

This chapter is about the number 3. In Roman numerals it is written III, in binary it is 11, and in base eight it is 3. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

3 is a prime number. Its only divisors are 1 and 3. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 3 counters in equal rows, the only way is a single row of 3. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 3 is one of them.

The sum of the divisors of 3 that are smaller than 3 is 1. Because this sum is smaller than 3, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 3 is 9 and its cube is 27. 3 is not a perfect square: it lies between 1 and 4. The sum of its digits is 3, so 3 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 3 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 3: find two whole numbers whose product is 6, then two whole numbers whose sum is 10. Then write 3 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 3 is even. A hint: divisors usually come in pairs whose product is 3, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 3 that has the same number of divisors, or test whether 4 and 2 share any common factor with 3. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 4. The number 4

This chapter is about the number 4. In Roman numerals it is written IV, in binary it is 100, and in base eight it is 4. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

4 is a composite number. Its divisors are 1, 2, 4, so it has 3 divisors in total. As a product of prime numbers it is written 2 × 2. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 4 that are smaller than 4 is 3. Because this sum is smaller than 4, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 4 is 16 and its cube is 64. 4 is itself a perfect square, because 2 × 2 = 4. The sum of its digits is 4, so 4 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 4 is even, so it can be shared equally between two people.

Exercise for chapter 4: find two whole numbers whose product is 8, then two whole numbers whose sum is 11. Then write 4 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 4 is odd. A hint: divisors usually come in pairs whose product is 4, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 4 that has the same number of divisors, or test whether 5 and 3 share any common factor with 4. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 5. The number 5

This chapter is about the number 5. In Roman numerals it is written V, in binary it is 101, and in base eight it is 5. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

5 is a prime number. Its only divisors are 1 and 5. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 5 counters in equal rows, the only way is a single row of 5. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 5 is one of them.

The sum of the divisors of 5 that are smaller than 5 is 1. Because this sum is smaller than 5, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 5 is 25 and its cube is 125. 5 is not a perfect square: it lies between 4 and 9. The sum of its digits is 5, so 5 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 5 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 5: find two whole numbers whose product is 10, then two whole numbers whose sum is 12. Then write 5 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 5 is even. A hint: divisors usually come in pairs whose product is 5, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 5 that has the same number of divisors, or test whether 6 and 4 share any common factor with 5. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 6. The number 6

This chapter is about the number 6. In Roman numerals it is written VI, in binary it is 110, and in base eight it is 6. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

6 is a composite number. Its divisors are 1, 2, 3, 6, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 6 that are smaller than 6 is 6. Because this sum is equal to 6, the number is called perfect. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 6 is 36 and its cube is 216. 6 is not a perfect square: it lies between 4 and 9. The sum of its digits is 6, so 6 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 6 is even, so it can be shared equally between two people.

Exercise for chapter 6: find two whole numbers whose product is 12, then two whole numbers whose sum is 13. Then write 6 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 6 is even. A hint: divisors usually come in pairs whose product is 6, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 6 that has the same number of divisors, or test whether 7 and 5 share any common factor with 6. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 7. The number 7

This chapter is about the number 7. In Roman numerals it is written VII, in binary it is 111, and in base eight it is 7. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

7 is a prime number. Its only divisors are 1 and 7. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 7 counters in equal rows, the only way is a single row of 7. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 7 is one of them.

The sum of the divisors of 7 that are smaller than 7 is 1. Because this sum is smaller than 7, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 7 is 49 and its cube is 343. 7 is not a perfect square: it lies between 4 and 9. The sum of its digits is 7, so 7 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 7 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 7: find two whole numbers whose product is 14, then two whole numbers whose sum is 14. Then write 7 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 7 is even. A hint: divisors usually come in pairs whose product is 7, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 7 that has the same number of divisors, or test whether 8 and 6 share any common factor with 7. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 8. The number 8

This chapter is about the number 8. In Roman numerals it is written VIII, in binary it is 1000, and in base eight it is 10. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

8 is a composite number. Its divisors are 1, 2, 4, 8, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 2 × 2. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 8 that are smaller than 8 is 7. Because this sum is smaller than 8, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 8 is 64 and its cube is 512. 8 is not a perfect square: it lies between 4 and 9. The sum of its digits is 8, so 8 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 8 is even, so it can be shared equally between two people.

Exercise for chapter 8: find two whole numbers whose product is 16, then two whole numbers whose sum is 15. Then write 8 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 8 is even. A hint: divisors usually come in pairs whose product is 8, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 8 that has the same number of divisors, or test whether 9 and 7 share any common factor with 8. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 9. The number 9

This chapter is about the number 9. In Roman numerals it is written IX, in binary it is 1001, and in base eight it is 11. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

9 is a composite number. Its divisors are 1, 3, 9, so it has 3 divisors in total. As a product of prime numbers it is written 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 9 that are smaller than 9 is 4. Because this sum is smaller than 9, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 9 is 81 and its cube is 729. 9 is itself a perfect square, because 3 × 3 = 9. The sum of its digits is 9, so 9 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 9 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 9: find two whole numbers whose product is 18, then two whole numbers whose sum is 16. Then write 9 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 9 is odd. A hint: divisors usually come in pairs whose product is 9, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 9 that has the same number of divisors, or test whether 10 and 8 share any common factor with 9. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 10. The number 10

This chapter is about the number 10. In Roman numerals it is written X, in binary it is 1010, and in base eight it is 12. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

10 is a composite number. Its divisors are 1, 2, 5, 10, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 10 that are smaller than 10 is 8. Because this sum is smaller than 10, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 10 is 100 and its cube is 1000. 10 is not a perfect square: it lies between 9 and 16. The sum of its digits is 1, so 10 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 10 is even, so it can be shared equally between two people.

Exercise for chapter 10: find two whole numbers whose product is 20, then two whole numbers whose sum is 17. Then write 10 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 10 is even. A hint: divisors usually come in pairs whose product is 10, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 10 that has the same number of divisors, or test whether 11 and 9 share any common factor with 10. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 11. The number 11

This chapter is about the number 11. In Roman numerals it is written XI, in binary it is 1011, and in base eight it is 13. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

11 is a prime number. Its only divisors are 1 and 11. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 11 counters in equal rows, the only way is a single row of 11. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 11 is one of them.

The sum of the divisors of 11 that are smaller than 11 is 1. Because this sum is smaller than 11, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 11 is 121 and its cube is 1331. 11 is not a perfect square: it lies between 9 and 16. The sum of its digits is 2, so 11 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 11 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 11: find two whole numbers whose product is 22, then two whole numbers whose sum is 18. Then write 11 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 11 is even. A hint: divisors usually come in pairs whose product is 11, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 11 that has the same number of divisors, or test whether 12 and 10 share any common factor with 11. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 12. The number 12

This chapter is about the number 12. In Roman numerals it is written XII, in binary it is 1100, and in base eight it is 14. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

12 is a composite number. Its divisors are 1, 2, 3, 4, 6, 12, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 12 that are smaller than 12 is 16. Because this sum is larger than 12, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 12 is 144 and its cube is 1728. 12 is not a perfect square: it lies between 9 and 16. The sum of its digits is 3, so 12 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 12 is even, so it can be shared equally between two people.

Exercise for chapter 12: find two whole numbers whose product is 24, then two whole numbers whose sum is 19. Then write 12 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 12 is even. A hint: divisors usually come in pairs whose product is 12, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 12 that has the same number of divisors, or test whether 13 and 11 share any common factor with 12. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 13. The number 13

This chapter is about the number 13. In Roman numerals it is written XIII, in binary it is 1101, and in base eight it is 15. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

13 is a prime number. Its only divisors are 1 and 13. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 13 counters in equal rows, the only way is a single row of 13. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 13 is one of them.

The sum of the divisors of 13 that are smaller than 13 is 1. Because this sum is smaller than 13, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 13 is 169 and its cube is 2197. 13 is not a perfect square: it lies between 9 and 16. The sum of its digits is 4, so 13 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 13 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 13: find two whole numbers whose product is 26, then two whole numbers whose sum is 20. Then write 13 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 13 is even. A hint: divisors usually come in pairs whose product is 13, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 13 that has the same number of divisors, or test whether 14 and 12 share any common factor with 13. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 14. The number 14

This chapter is about the number 14. In Roman numerals it is written XIV, in binary it is 1110, and in base eight it is 16. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

14 is a composite number. Its divisors are 1, 2, 7, 14, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 14 that are smaller than 14 is 10. Because this sum is smaller than 14, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 14 is 196 and its cube is 2744. 14 is not a perfect square: it lies between 9 and 16. The sum of its digits is 5, so 14 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 14 is even, so it can be shared equally between two people.

Exercise for chapter 14: find two whole numbers whose product is 28, then two whole numbers whose sum is 21. Then write 14 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 14 is even. A hint: divisors usually come in pairs whose product is 14, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 14 that has the same number of divisors, or test whether 15 and 13 share any common factor with 14. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 15. The number 15

This chapter is about the number 15. In Roman numerals it is written XV, in binary it is 1111, and in base eight it is 17. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

15 is a composite number. Its divisors are 1, 3, 5, 15, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 15 that are smaller than 15 is 9. Because this sum is smaller than 15, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 15 is 225 and its cube is 3375. 15 is not a perfect square: it lies between 9 and 16. The sum of its digits is 6, so 15 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 15 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 15: find two whole numbers whose product is 30, then two whole numbers whose sum is 22. Then write 15 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 15 is even. A hint: divisors usually come in pairs whose product is 15, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 15 that has the same number of divisors, or test whether 16 and 14 share any common factor with 15. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 16. The number 16

This chapter is about the number 16. In Roman numerals it is written XVI, in binary it is 10000, and in base eight it is 20. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

16 is a composite number. Its divisors are 1, 2, 4, 8, 16, so it has 5 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 16 that are smaller than 16 is 15. Because this sum is smaller than 16, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 16 is 256 and its cube is 4096. 16 is itself a perfect square, because 4 × 4 = 16. The sum of its digits is 7, so 16 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 16 is even, so it can be shared equally between two people.

Exercise for chapter 16: find two whole numbers whose product is 32, then two whole numbers whose sum is 23. Then write 16 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 16 is odd. A hint: divisors usually come in pairs whose product is 16, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 16 that has the same number of divisors, or test whether 17 and 15 share any common factor with 16. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 17. The number 17

This chapter is about the number 17. In Roman numerals it is written XVII, in binary it is 10001, and in base eight it is 21. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

17 is a prime number. Its only divisors are 1 and 17. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 17 counters in equal rows, the only way is a single row of 17. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 17 is one of them.

The sum of the divisors of 17 that are smaller than 17 is 1. Because this sum is smaller than 17, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 17 is 289 and its cube is 4913. 17 is not a perfect square: it lies between 16 and 25. The sum of its digits is 8, so 17 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 17 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 17: find two whole numbers whose product is 34, then two whole numbers whose sum is 24. Then write 17 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 17 is even. A hint: divisors usually come in pairs whose product is 17, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 17 that has the same number of divisors, or test whether 18 and 16 share any common factor with 17. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 18. The number 18

This chapter is about the number 18. In Roman numerals it is written XVIII, in binary it is 10010, and in base eight it is 22. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

18 is a composite number. Its divisors are 1, 2, 3, 6, 9, 18, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 18 that are smaller than 18 is 21. Because this sum is larger than 18, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 18 is 324 and its cube is 5832. 18 is not a perfect square: it lies between 16 and 25. The sum of its digits is 9, so 18 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 18 is even, so it can be shared equally between two people.

Exercise for chapter 18: find two whole numbers whose product is 36, then two whole numbers whose sum is 25. Then write 18 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 18 is even. A hint: divisors usually come in pairs whose product is 18, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 18 that has the same number of divisors, or test whether 19 and 17 share any common factor with 18. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 19. The number 19

This chapter is about the number 19. In Roman numerals it is written XIX, in binary it is 10011, and in base eight it is 23. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

19 is a prime number. Its only divisors are 1 and 19. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 19 counters in equal rows, the only way is a single row of 19. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 19 is one of them.

The sum of the divisors of 19 that are smaller than 19 is 1. Because this sum is smaller than 19, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 19 is 361 and its cube is 6859. 19 is not a perfect square: it lies between 16 and 25. The sum of its digits is 10, so 19 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 19 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 19: find two whole numbers whose product is 38, then two whole numbers whose sum is 26. Then write 19 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 19 is even. A hint: divisors usually come in pairs whose product is 19, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 19 that has the same number of divisors, or test whether 20 and 18 share any common factor with 19. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 20. The number 20

This chapter is about the number 20. In Roman numerals it is written XX, in binary it is 10100, and in base eight it is 24. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

20 is a composite number. Its divisors are 1, 2, 4, 5, 10, 20, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 20 that are smaller than 20 is 22. Because this sum is larger than 20, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 20 is 400 and its cube is 8000. 20 is not a perfect square: it lies between 16 and 25. The sum of its digits is 2, so 20 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 20 is even, so it can be shared equally between two people.

Exercise for chapter 20: find two whole numbers whose product is 40, then two whole numbers whose sum is 27. Then write 20 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 20 is even. A hint: divisors usually come in pairs whose product is 20, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 20 that has the same number of divisors, or test whether 21 and 19 share any common factor with 20. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 21. The number 21

This chapter is about the number 21. In Roman numerals it is written XXI, in binary it is 10101, and in base eight it is 25. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

21 is a composite number. Its divisors are 1, 3, 7, 21, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 21 that are smaller than 21 is 11. Because this sum is smaller than 21, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 21 is 441 and its cube is 9261. 21 is not a perfect square: it lies between 16 and 25. The sum of its digits is 3, so 21 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 21 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 21: find two whole numbers whose product is 42, then two whole numbers whose sum is 28. Then write 21 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 21 is even. A hint: divisors usually come in pairs whose product is 21, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 21 that has the same number of divisors, or test whether 22 and 20 share any common factor with 21. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 22. The number 22

This chapter is about the number 22. In Roman numerals it is written XXII, in binary it is 10110, and in base eight it is 26. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

22 is a composite number. Its divisors are 1, 2, 11, 22, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 22 that are smaller than 22 is 14. Because this sum is smaller than 22, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 22 is 484 and its cube is 10648. 22 is not a perfect square: it lies between 16 and 25. The sum of its digits is 4, so 22 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 22 is even, so it can be shared equally between two people.

Exercise for chapter 22: find two whole numbers whose product is 44, then two whole numbers whose sum is 29. Then write 22 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 22 is even. A hint: divisors usually come in pairs whose product is 22, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 22 that has the same number of divisors, or test whether 23 and 21 share any common factor with 22. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 23. The number 23

This chapter is about the number 23. In Roman numerals it is written XXIII, in binary it is 10111, and in base eight it is 27. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

23 is a prime number. Its only divisors are 1 and 23. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 23 counters in equal rows, the only way is a single row of 23. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 23 is one of them.

The sum of the divisors of 23 that are smaller than 23 is 1. Because this sum is smaller than 23, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 23 is 529 and its cube is 12167. 23 is not a perfect square: it lies between 16 and 25. The sum of its digits is 5, so 23 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 23 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 23: find two whole numbers whose product is 46, then two whole numbers whose sum is 30. Then write 23 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 23 is even. A hint: divisors usually come in pairs whose product is 23, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 23 that has the same number of divisors, or test whether 24 and 22 share any common factor with 23. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 24. The number 24

This chapter is about the number 24. In Roman numerals it is written XXIV, in binary it is 11000, and in base eight it is 30. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

24 is a composite number. Its divisors are 1, 2, 3, 4, 6, 8, 12, 24, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 24 that are smaller than 24 is 36. Because this sum is larger than 24, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 24 is 576 and its cube is 13824. 24 is not a perfect square: it lies between 16 and 25. The sum of its digits is 6, so 24 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 24 is even, so it can be shared equally between two people.

Exercise for chapter 24: find two whole numbers whose product is 48, then two whole numbers whose sum is 31. Then write 24 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 24 is even. A hint: divisors usually come in pairs whose product is 24, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 24 that has the same number of divisors, or test whether 25 and 23 share any common factor with 24. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 25. The number 25

This chapter is about the number 25. In Roman numerals it is written XXV, in binary it is 11001, and in base eight it is 31. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

25 is a composite number. Its divisors are 1, 5, 25, so it has 3 divisors in total. As a product of prime numbers it is written 5 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 25 that are smaller than 25 is 6. Because this sum is smaller than 25, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 25 is 625 and its cube is 15625. 25 is itself a perfect square, because 5 × 5 = 25. The sum of its digits is 7, so 25 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 25 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 25: find two whole numbers whose product is 50, then two whole numbers whose sum is 32. Then write 25 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 25 is odd. A hint: divisors usually come in pairs whose product is 25, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 25 that has the same number of divisors, or test whether 26 and 24 share any common factor with 25. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 26. The number 26

This chapter is about the number 26. In Roman numerals it is written XXVI, in binary it is 11010, and in base eight it is 32. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

26 is a composite number. Its divisors are 1, 2, 13, 26, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 26 that are smaller than 26 is 16. Because this sum is smaller than 26, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 26 is 676 and its cube is 17576. 26 is not a perfect square: it lies between 25 and 36. The sum of its digits is 8, so 26 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 26 is even, so it can be shared equally between two people.

Exercise for chapter 26: find two whole numbers whose product is 52, then two whole numbers whose sum is 33. Then write 26 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 26 is even. A hint: divisors usually come in pairs whose product is 26, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 26 that has the same number of divisors, or test whether 27 and 25 share any common factor with 26. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 27. The number 27

This chapter is about the number 27. In Roman numerals it is written XXVII, in binary it is 11011, and in base eight it is 33. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

27 is a composite number. Its divisors are 1, 3, 9, 27, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 27 that are smaller than 27 is 13. Because this sum is smaller than 27, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 27 is 729 and its cube is 19683. 27 is not a perfect square: it lies between 25 and 36. The sum of its digits is 9, so 27 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 27 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 27: find two whole numbers whose product is 54, then two whole numbers whose sum is 34. Then write 27 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 27 is even. A hint: divisors usually come in pairs whose product is 27, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 27 that has the same number of divisors, or test whether 28 and 26 share any common factor with 27. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 28. The number 28

This chapter is about the number 28. In Roman numerals it is written XXVIII, in binary it is 11100, and in base eight it is 34. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

28 is a composite number. Its divisors are 1, 2, 4, 7, 14, 28, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 28 that are smaller than 28 is 28. Because this sum is equal to 28, the number is called perfect. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 28 is 784 and its cube is 21952. 28 is not a perfect square: it lies between 25 and 36. The sum of its digits is 10, so 28 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 28 is even, so it can be shared equally between two people.

Exercise for chapter 28: find two whole numbers whose product is 56, then two whole numbers whose sum is 35. Then write 28 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 28 is even. A hint: divisors usually come in pairs whose product is 28, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 28 that has the same number of divisors, or test whether 29 and 27 share any common factor with 28. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 29. The number 29

This chapter is about the number 29. In Roman numerals it is written XXIX, in binary it is 11101, and in base eight it is 35. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

29 is a prime number. Its only divisors are 1 and 29. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 29 counters in equal rows, the only way is a single row of 29. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 29 is one of them.

The sum of the divisors of 29 that are smaller than 29 is 1. Because this sum is smaller than 29, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 29 is 841 and its cube is 24389. 29 is not a perfect square: it lies between 25 and 36. The sum of its digits is 11, so 29 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 29 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 29: find two whole numbers whose product is 58, then two whole numbers whose sum is 36. Then write 29 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 29 is even. A hint: divisors usually come in pairs whose product is 29, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 29 that has the same number of divisors, or test whether 30 and 28 share any common factor with 29. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 30. The number 30

This chapter is about the number 30. In Roman numerals it is written XXX, in binary it is 11110, and in base eight it is 36. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

30 is a composite number. Its divisors are 1, 2, 3, 5, 6, 10, 15, 30, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 30 that are smaller than 30 is 42. Because this sum is larger than 30, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 30 is 900 and its cube is 27000. 30 is not a perfect square: it lies between 25 and 36. The sum of its digits is 3, so 30 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 30 is even, so it can be shared equally between two people.

Exercise for chapter 30: find two whole numbers whose product is 60, then two whole numbers whose sum is 37. Then write 30 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 30 is even. A hint: divisors usually come in pairs whose product is 30, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 30 that has the same number of divisors, or test whether 31 and 29 share any common factor with 30. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 31. The number 31

This chapter is about the number 31. In Roman numerals it is written XXXI, in binary it is 11111, and in base eight it is 37. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

31 is a prime number. Its only divisors are 1 and 31. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 31 counters in equal rows, the only way is a single row of 31. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 31 is one of them.

The sum of the divisors of 31 that are smaller than 31 is 1. Because this sum is smaller than 31, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 31 is 961 and its cube is 29791. 31 is not a perfect square: it lies between 25 and 36. The sum of its digits is 4, so 31 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 31 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 31: find two whole numbers whose product is 62, then two whole numbers whose sum is 38. Then write 31 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 31 is even. A hint: divisors usually come in pairs whose product is 31, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 31 that has the same number of divisors, or test whether 32 and 30 share any common factor with 31. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 32. The number 32

This chapter is about the number 32. In Roman numerals it is written XXXII, in binary it is 100000, and in base eight it is 40. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

32 is a composite number. Its divisors are 1, 2, 4, 8, 16, 32, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 2. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 32 that are smaller than 32 is 31. Because this sum is smaller than 32, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 32 is 1024 and its cube is 32768. 32 is not a perfect square: it lies between 25 and 36. The sum of its digits is 5, so 32 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 32 is even, so it can be shared equally between two people.

Exercise for chapter 32: find two whole numbers whose product is 64, then two whole numbers whose sum is 39. Then write 32 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 32 is even. A hint: divisors usually come in pairs whose product is 32, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 32 that has the same number of divisors, or test whether 33 and 31 share any common factor with 32. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 33. The number 33

This chapter is about the number 33. In Roman numerals it is written XXXIII, in binary it is 100001, and in base eight it is 41. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

33 is a composite number. Its divisors are 1, 3, 11, 33, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 33 that are smaller than 33 is 15. Because this sum is smaller than 33, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 33 is 1089 and its cube is 35937. 33 is not a perfect square: it lies between 25 and 36. The sum of its digits is 6, so 33 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 33 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 33: find two whole numbers whose product is 66, then two whole numbers whose sum is 40. Then write 33 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 33 is even. A hint: divisors usually come in pairs whose product is 33, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 33 that has the same number of divisors, or test whether 34 and 32 share any common factor with 33. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 34. The number 34

This chapter is about the number 34. In Roman numerals it is written XXXIV, in binary it is 100010, and in base eight it is 42. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

34 is a composite number. Its divisors are 1, 2, 17, 34, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 34 that are smaller than 34 is 20. Because this sum is smaller than 34, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 34 is 1156 and its cube is 39304. 34 is not a perfect square: it lies between 25 and 36. The sum of its digits is 7, so 34 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 34 is even, so it can be shared equally between two people.

Exercise for chapter 34: find two whole numbers whose product is 68, then two whole numbers whose sum is 41. Then write 34 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 34 is even. A hint: divisors usually come in pairs whose product is 34, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 34 that has the same number of divisors, or test whether 35 and 33 share any common factor with 34. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 35. The number 35

This chapter is about the number 35. In Roman numerals it is written XXXV, in binary it is 100011, and in base eight it is 43. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

35 is a composite number. Its divisors are 1, 5, 7, 35, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 35 that are smaller than 35 is 13. Because this sum is smaller than 35, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 35 is 1225 and its cube is 42875. 35 is not a perfect square: it lies between 25 and 36. The sum of its digits is 8, so 35 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 35 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 35: find two whole numbers whose product is 70, then two whole numbers whose sum is 42. Then write 35 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 35 is even. A hint: divisors usually come in pairs whose product is 35, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 35 that has the same number of divisors, or test whether 36 and 34 share any common factor with 35. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 36. The number 36

This chapter is about the number 36. In Roman numerals it is written XXXVI, in binary it is 100100, and in base eight it is 44. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

36 is a composite number. Its divisors are 1, 2, 3, 4, 6, 9, 12, 18, 36, so it has 9 divisors in total. As a product of prime numbers it is written 2 × 2 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 36 that are smaller than 36 is 55. Because this sum is larger than 36, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 36 is 1296 and its cube is 46656. 36 is itself a perfect square, because 6 × 6 = 36. The sum of its digits is 9, so 36 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 36 is even, so it can be shared equally between two people.

Exercise for chapter 36: find two whole numbers whose product is 72, then two whole numbers whose sum is 43. Then write 36 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 36 is odd. A hint: divisors usually come in pairs whose product is 36, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 36 that has the same number of divisors, or test whether 37 and 35 share any common factor with 36. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 37. The number 37

This chapter is about the number 37. In Roman numerals it is written XXXVII, in binary it is 100101, and in base eight it is 45. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

37 is a prime number. Its only divisors are 1 and 37. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 37 counters in equal rows, the only way is a single row of 37. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 37 is one of them.

The sum of the divisors of 37 that are smaller than 37 is 1. Because this sum is smaller than 37, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 37 is 1369 and its cube is 50653. 37 is not a perfect square: it lies between 36 and 49. The sum of its digits is 10, so 37 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 37 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 37: find two whole numbers whose product is 74, then two whole numbers whose sum is 44. Then write 37 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 37 is even. A hint: divisors usually come in pairs whose product is 37, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 37 that has the same number of divisors, or test whether 38 and 36 share any common factor with 37. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 38. The number 38

This chapter is about the number 38. In Roman numerals it is written XXXVIII, in binary it is 100110, and in base eight it is 46. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

38 is a composite number. Its divisors are 1, 2, 19, 38, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 38 that are smaller than 38 is 22. Because this sum is smaller than 38, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 38 is 1444 and its cube is 54872. 38 is not a perfect square: it lies between 36 and 49. The sum of its digits is 11, so 38 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 38 is even, so it can be shared equally between two people.

Exercise for chapter 38: find two whole numbers whose product is 76, then two whole numbers whose sum is 45. Then write 38 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 38 is even. A hint: divisors usually come in pairs whose product is 38, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 38 that has the same number of divisors, or test whether 39 and 37 share any common factor with 38. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 39. The number 39

This chapter is about the number 39. In Roman numerals it is written XXXIX, in binary it is 100111, and in base eight it is 47. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

39 is a composite number. Its divisors are 1, 3, 13, 39, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 39 that are smaller than 39 is 17. Because this sum is smaller than 39, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 39 is 1521 and its cube is 59319. 39 is not a perfect square: it lies between 36 and 49. The sum of its digits is 12, so 39 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 39 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 39: find two whole numbers whose product is 78, then two whole numbers whose sum is 46. Then write 39 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 39 is even. A hint: divisors usually come in pairs whose product is 39, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 39 that has the same number of divisors, or test whether 40 and 38 share any common factor with 39. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 40. The number 40

This chapter is about the number 40. In Roman numerals it is written XL, in binary it is 101000, and in base eight it is 50. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

40 is a composite number. Its divisors are 1, 2, 4, 5, 8, 10, 20, 40, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 40 that are smaller than 40 is 50. Because this sum is larger than 40, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 40 is 1600 and its cube is 64000. 40 is not a perfect square: it lies between 36 and 49. The sum of its digits is 4, so 40 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 40 is even, so it can be shared equally between two people.

Exercise for chapter 40: find two whole numbers whose product is 80, then two whole numbers whose sum is 47. Then write 40 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 40 is even. A hint: divisors usually come in pairs whose product is 40, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 40 that has the same number of divisors, or test whether 41 and 39 share any common factor with 40. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 41. The number 41

This chapter is about the number 41. In Roman numerals it is written XLI, in binary it is 101001, and in base eight it is 51. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

41 is a prime number. Its only divisors are 1 and 41. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 41 counters in equal rows, the only way is a single row of 41. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 41 is one of them.

The sum of the divisors of 41 that are smaller than 41 is 1. Because this sum is smaller than 41, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 41 is 1681 and its cube is 68921. 41 is not a perfect square: it lies between 36 and 49. The sum of its digits is 5, so 41 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 41 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 41: find two whole numbers whose product is 82, then two whole numbers whose sum is 48. Then write 41 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 41 is even. A hint: divisors usually come in pairs whose product is 41, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 41 that has the same number of divisors, or test whether 42 and 40 share any common factor with 41. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 42. The number 42

This chapter is about the number 42. In Roman numerals it is written XLII, in binary it is 101010, and in base eight it is 52. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

42 is a composite number. Its divisors are 1, 2, 3, 6, 7, 14, 21, 42, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 42 that are smaller than 42 is 54. Because this sum is larger than 42, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 42 is 1764 and its cube is 74088. 42 is not a perfect square: it lies between 36 and 49. The sum of its digits is 6, so 42 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 42 is even, so it can be shared equally between two people.

Exercise for chapter 42: find two whole numbers whose product is 84, then two whole numbers whose sum is 49. Then write 42 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 42 is even. A hint: divisors usually come in pairs whose product is 42, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 42 that has the same number of divisors, or test whether 43 and 41 share any common factor with 42. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 43. The number 43

This chapter is about the number 43. In Roman numerals it is written XLIII, in binary it is 101011, and in base eight it is 53. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

43 is a prime number. Its only divisors are 1 and 43. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 43 counters in equal rows, the only way is a single row of 43. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 43 is one of them.

The sum of the divisors of 43 that are smaller than 43 is 1. Because this sum is smaller than 43, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 43 is 1849 and its cube is 79507. 43 is not a perfect square: it lies between 36 and 49. The sum of its digits is 7, so 43 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 43 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 43: find two whole numbers whose product is 86, then two whole numbers whose sum is 50. Then write 43 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 43 is even. A hint: divisors usually come in pairs whose product is 43, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 43 that has the same number of divisors, or test whether 44 and 42 share any common factor with 43. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 44. The number 44

This chapter is about the number 44. In Roman numerals it is written XLIV, in binary it is 101100, and in base eight it is 54. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

44 is a composite number. Its divisors are 1, 2, 4, 11, 22, 44, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 44 that are smaller than 44 is 40. Because this sum is smaller than 44, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 44 is 1936 and its cube is 85184. 44 is not a perfect square: it lies between 36 and 49. The sum of its digits is 8, so 44 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 44 is even, so it can be shared equally between two people.

Exercise for chapter 44: find two whole numbers whose product is 88, then two whole numbers whose sum is 51. Then write 44 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 44 is even. A hint: divisors usually come in pairs whose product is 44, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 44 that has the same number of divisors, or test whether 45 and 43 share any common factor with 44. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 45. The number 45

This chapter is about the number 45. In Roman numerals it is written XLV, in binary it is 101101, and in base eight it is 55. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

45 is a composite number. Its divisors are 1, 3, 5, 9, 15, 45, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 45 that are smaller than 45 is 33. Because this sum is smaller than 45, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 45 is 2025 and its cube is 91125. 45 is not a perfect square: it lies between 36 and 49. The sum of its digits is 9, so 45 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 45 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 45: find two whole numbers whose product is 90, then two whole numbers whose sum is 52. Then write 45 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 45 is even. A hint: divisors usually come in pairs whose product is 45, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 45 that has the same number of divisors, or test whether 46 and 44 share any common factor with 45. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 46. The number 46

This chapter is about the number 46. In Roman numerals it is written XLVI, in binary it is 101110, and in base eight it is 56. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

46 is a composite number. Its divisors are 1, 2, 23, 46, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 23. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 46 that are smaller than 46 is 26. Because this sum is smaller than 46, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 46 is 2116 and its cube is 97336. 46 is not a perfect square: it lies between 36 and 49. The sum of its digits is 10, so 46 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 46 is even, so it can be shared equally between two people.

Exercise for chapter 46: find two whole numbers whose product is 92, then two whole numbers whose sum is 53. Then write 46 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 46 is even. A hint: divisors usually come in pairs whose product is 46, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 46 that has the same number of divisors, or test whether 47 and 45 share any common factor with 46. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 47. The number 47

This chapter is about the number 47. In Roman numerals it is written XLVII, in binary it is 101111, and in base eight it is 57. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

47 is a prime number. Its only divisors are 1 and 47. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 47 counters in equal rows, the only way is a single row of 47. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 47 is one of them.

The sum of the divisors of 47 that are smaller than 47 is 1. Because this sum is smaller than 47, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 47 is 2209 and its cube is 103823. 47 is not a perfect square: it lies between 36 and 49. The sum of its digits is 11, so 47 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 47 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 47: find two whole numbers whose product is 94, then two whole numbers whose sum is 54. Then write 47 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 47 is even. A hint: divisors usually come in pairs whose product is 47, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 47 that has the same number of divisors, or test whether 48 and 46 share any common factor with 47. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 48. The number 48

This chapter is about the number 48. In Roman numerals it is written XLVIII, in binary it is 110000, and in base eight it is 60. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

48 is a composite number. Its divisors are 1, 2, 3, 4, 6, 8, 12, 16, 24, 48, so it has 10 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 48 that are smaller than 48 is 76. Because this sum is larger than 48, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 48 is 2304 and its cube is 110592. 48 is not a perfect square: it lies between 36 and 49. The sum of its digits is 12, so 48 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 48 is even, so it can be shared equally between two people.

Exercise for chapter 48: find two whole numbers whose product is 96, then two whole numbers whose sum is 55. Then write 48 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 48 is even. A hint: divisors usually come in pairs whose product is 48, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 48 that has the same number of divisors, or test whether 49 and 47 share any common factor with 48. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 49. The number 49

This chapter is about the number 49. In Roman numerals it is written XLIX, in binary it is 110001, and in base eight it is 61. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

49 is a composite number. Its divisors are 1, 7, 49, so it has 3 divisors in total. As a product of prime numbers it is written 7 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 49 that are smaller than 49 is 8. Because this sum is smaller than 49, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 49 is 2401 and its cube is 117649. 49 is itself a perfect square, because 7 × 7 = 49. The sum of its digits is 13, so 49 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 49 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 49: find two whole numbers whose product is 98, then two whole numbers whose sum is 56. Then write 49 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 49 is odd. A hint: divisors usually come in pairs whose product is 49, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 49 that has the same number of divisors, or test whether 50 and 48 share any common factor with 49. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 50. The number 50

This chapter is about the number 50. In Roman numerals it is written L, in binary it is 110010, and in base eight it is 62. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

50 is a composite number. Its divisors are 1, 2, 5, 10, 25, 50, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 5 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 50 that are smaller than 50 is 43. Because this sum is smaller than 50, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 50 is 2500 and its cube is 125000. 50 is not a perfect square: it lies between 49 and 64. The sum of its digits is 5, so 50 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 50 is even, so it can be shared equally between two people.

Exercise for chapter 50: find two whole numbers whose product is 100, then two whole numbers whose sum is 57. Then write 50 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 50 is even. A hint: divisors usually come in pairs whose product is 50, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 50 that has the same number of divisors, or test whether 51 and 49 share any common factor with 50. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 51. The number 51

This chapter is about the number 51. In Roman numerals it is written LI, in binary it is 110011, and in base eight it is 63. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

51 is a composite number. Its divisors are 1, 3, 17, 51, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 51 that are smaller than 51 is 21. Because this sum is smaller than 51, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 51 is 2601 and its cube is 132651. 51 is not a perfect square: it lies between 49 and 64. The sum of its digits is 6, so 51 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 51 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 51: find two whole numbers whose product is 102, then two whole numbers whose sum is 58. Then write 51 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 51 is even. A hint: divisors usually come in pairs whose product is 51, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 51 that has the same number of divisors, or test whether 52 and 50 share any common factor with 51. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 52. The number 52

This chapter is about the number 52. In Roman numerals it is written LII, in binary it is 110100, and in base eight it is 64. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

52 is a composite number. Its divisors are 1, 2, 4, 13, 26, 52, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 52 that are smaller than 52 is 46. Because this sum is smaller than 52, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 52 is 2704 and its cube is 140608. 52 is not a perfect square: it lies between 49 and 64. The sum of its digits is 7, so 52 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 52 is even, so it can be shared equally between two people.

Exercise for chapter 52: find two whole numbers whose product is 104, then two whole numbers whose sum is 59. Then write 52 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 52 is even. A hint: divisors usually come in pairs whose product is 52, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 52 that has the same number of divisors, or test whether 53 and 51 share any common factor with 52. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 53. The number 53

This chapter is about the number 53. In Roman numerals it is written LIII, in binary it is 110101, and in base eight it is 65. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

53 is a prime number. Its only divisors are 1 and 53. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 53 counters in equal rows, the only way is a single row of 53. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 53 is one of them.

The sum of the divisors of 53 that are smaller than 53 is 1. Because this sum is smaller than 53, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 53 is 2809 and its cube is 148877. 53 is not a perfect square: it lies between 49 and 64. The sum of its digits is 8, so 53 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 53 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 53: find two whole numbers whose product is 106, then two whole numbers whose sum is 60. Then write 53 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 53 is even. A hint: divisors usually come in pairs whose product is 53, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 53 that has the same number of divisors, or test whether 54 and 52 share any common factor with 53. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 54. The number 54

This chapter is about the number 54. In Roman numerals it is written LIV, in binary it is 110110, and in base eight it is 66. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

54 is a composite number. Its divisors are 1, 2, 3, 6, 9, 18, 27, 54, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 54 that are smaller than 54 is 66. Because this sum is larger than 54, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 54 is 2916 and its cube is 157464. 54 is not a perfect square: it lies between 49 and 64. The sum of its digits is 9, so 54 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 54 is even, so it can be shared equally between two people.

Exercise for chapter 54: find two whole numbers whose product is 108, then two whole numbers whose sum is 61. Then write 54 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 54 is even. A hint: divisors usually come in pairs whose product is 54, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 54 that has the same number of divisors, or test whether 55 and 53 share any common factor with 54. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 55. The number 55

This chapter is about the number 55. In Roman numerals it is written LV, in binary it is 110111, and in base eight it is 67. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

55 is a composite number. Its divisors are 1, 5, 11, 55, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 55 that are smaller than 55 is 17. Because this sum is smaller than 55, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 55 is 3025 and its cube is 166375. 55 is not a perfect square: it lies between 49 and 64. The sum of its digits is 10, so 55 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 55 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 55: find two whole numbers whose product is 110, then two whole numbers whose sum is 62. Then write 55 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 55 is even. A hint: divisors usually come in pairs whose product is 55, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 55 that has the same number of divisors, or test whether 56 and 54 share any common factor with 55. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 56. The number 56

This chapter is about the number 56. In Roman numerals it is written LVI, in binary it is 111000, and in base eight it is 70. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

56 is a composite number. Its divisors are 1, 2, 4, 7, 8, 14, 28, 56, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 56 that are smaller than 56 is 64. Because this sum is larger than 56, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 56 is 3136 and its cube is 175616. 56 is not a perfect square: it lies between 49 and 64. The sum of its digits is 11, so 56 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 56 is even, so it can be shared equally between two people.

Exercise for chapter 56: find two whole numbers whose product is 112, then two whole numbers whose sum is 63. Then write 56 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 56 is even. A hint: divisors usually come in pairs whose product is 56, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 56 that has the same number of divisors, or test whether 57 and 55 share any common factor with 56. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 57. The number 57

This chapter is about the number 57. In Roman numerals it is written LVII, in binary it is 111001, and in base eight it is 71. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

57 is a composite number. Its divisors are 1, 3, 19, 57, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 57 that are smaller than 57 is 23. Because this sum is smaller than 57, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 57 is 3249 and its cube is 185193. 57 is not a perfect square: it lies between 49 and 64. The sum of its digits is 12, so 57 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 57 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 57: find two whole numbers whose product is 114, then two whole numbers whose sum is 64. Then write 57 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 57 is even. A hint: divisors usually come in pairs whose product is 57, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 57 that has the same number of divisors, or test whether 58 and 56 share any common factor with 57. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 58. The number 58

This chapter is about the number 58. In Roman numerals it is written LVIII, in binary it is 111010, and in base eight it is 72. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

58 is a composite number. Its divisors are 1, 2, 29, 58, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 29. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 58 that are smaller than 58 is 32. Because this sum is smaller than 58, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 58 is 3364 and its cube is 195112. 58 is not a perfect square: it lies between 49 and 64. The sum of its digits is 13, so 58 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 58 is even, so it can be shared equally between two people.

Exercise for chapter 58: find two whole numbers whose product is 116, then two whole numbers whose sum is 65. Then write 58 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 58 is even. A hint: divisors usually come in pairs whose product is 58, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 58 that has the same number of divisors, or test whether 59 and 57 share any common factor with 58. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 59. The number 59

This chapter is about the number 59. In Roman numerals it is written LIX, in binary it is 111011, and in base eight it is 73. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

59 is a prime number. Its only divisors are 1 and 59. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 59 counters in equal rows, the only way is a single row of 59. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 59 is one of them.

The sum of the divisors of 59 that are smaller than 59 is 1. Because this sum is smaller than 59, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 59 is 3481 and its cube is 205379. 59 is not a perfect square: it lies between 49 and 64. The sum of its digits is 14, so 59 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 59 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 59: find two whole numbers whose product is 118, then two whole numbers whose sum is 66. Then write 59 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 59 is even. A hint: divisors usually come in pairs whose product is 59, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 59 that has the same number of divisors, or test whether 60 and 58 share any common factor with 59. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 60. The number 60

This chapter is about the number 60. In Roman numerals it is written LX, in binary it is 111100, and in base eight it is 74. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

60 is a composite number. Its divisors are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 60 that are smaller than 60 is 108. Because this sum is larger than 60, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 60 is 3600 and its cube is 216000. 60 is not a perfect square: it lies between 49 and 64. The sum of its digits is 6, so 60 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 60 is even, so it can be shared equally between two people.

Exercise for chapter 60: find two whole numbers whose product is 120, then two whole numbers whose sum is 67. Then write 60 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 60 is even. A hint: divisors usually come in pairs whose product is 60, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 60 that has the same number of divisors, or test whether 61 and 59 share any common factor with 60. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 61. The number 61

This chapter is about the number 61. In Roman numerals it is written LXI, in binary it is 111101, and in base eight it is 75. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

61 is a prime number. Its only divisors are 1 and 61. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 61 counters in equal rows, the only way is a single row of 61. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 61 is one of them.

The sum of the divisors of 61 that are smaller than 61 is 1. Because this sum is smaller than 61, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 61 is 3721 and its cube is 226981. 61 is not a perfect square: it lies between 49 and 64. The sum of its digits is 7, so 61 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 61 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 61: find two whole numbers whose product is 122, then two whole numbers whose sum is 68. Then write 61 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 61 is even. A hint: divisors usually come in pairs whose product is 61, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 61 that has the same number of divisors, or test whether 62 and 60 share any common factor with 61. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 62. The number 62

This chapter is about the number 62. In Roman numerals it is written LXII, in binary it is 111110, and in base eight it is 76. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

62 is a composite number. Its divisors are 1, 2, 31, 62, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 31. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 62 that are smaller than 62 is 34. Because this sum is smaller than 62, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 62 is 3844 and its cube is 238328. 62 is not a perfect square: it lies between 49 and 64. The sum of its digits is 8, so 62 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 62 is even, so it can be shared equally between two people.

Exercise for chapter 62: find two whole numbers whose product is 124, then two whole numbers whose sum is 69. Then write 62 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 62 is even. A hint: divisors usually come in pairs whose product is 62, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 62 that has the same number of divisors, or test whether 63 and 61 share any common factor with 62. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 63. The number 63

This chapter is about the number 63. In Roman numerals it is written LXIII, in binary it is 111111, and in base eight it is 77. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

63 is a composite number. Its divisors are 1, 3, 7, 9, 21, 63, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 3 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 63 that are smaller than 63 is 41. Because this sum is smaller than 63, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 63 is 3969 and its cube is 250047. 63 is not a perfect square: it lies between 49 and 64. The sum of its digits is 9, so 63 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 63 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 63: find two whole numbers whose product is 126, then two whole numbers whose sum is 70. Then write 63 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 63 is even. A hint: divisors usually come in pairs whose product is 63, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 63 that has the same number of divisors, or test whether 64 and 62 share any common factor with 63. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 64. The number 64

This chapter is about the number 64. In Roman numerals it is written LXIV, in binary it is 1000000, and in base eight it is 100. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

64 is a composite number. Its divisors are 1, 2, 4, 8, 16, 32, 64, so it has 7 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 2 × 2. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 64 that are smaller than 64 is 63. Because this sum is smaller than 64, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 64 is 4096 and its cube is 262144. 64 is itself a perfect square, because 8 × 8 = 64. The sum of its digits is 10, so 64 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 64 is even, so it can be shared equally between two people.

Exercise for chapter 64: find two whole numbers whose product is 128, then two whole numbers whose sum is 71. Then write 64 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 64 is odd. A hint: divisors usually come in pairs whose product is 64, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 64 that has the same number of divisors, or test whether 65 and 63 share any common factor with 64. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 65. The number 65

This chapter is about the number 65. In Roman numerals it is written LXV, in binary it is 1000001, and in base eight it is 101. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

65 is a composite number. Its divisors are 1, 5, 13, 65, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 65 that are smaller than 65 is 19. Because this sum is smaller than 65, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 65 is 4225 and its cube is 274625. 65 is not a perfect square: it lies between 64 and 81. The sum of its digits is 11, so 65 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 65 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 65: find two whole numbers whose product is 130, then two whole numbers whose sum is 72. Then write 65 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 65 is even. A hint: divisors usually come in pairs whose product is 65, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 65 that has the same number of divisors, or test whether 66 and 64 share any common factor with 65. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 66. The number 66

This chapter is about the number 66. In Roman numerals it is written LXVI, in binary it is 1000010, and in base eight it is 102. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

66 is a composite number. Its divisors are 1, 2, 3, 6, 11, 22, 33, 66, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 66 that are smaller than 66 is 78. Because this sum is larger than 66, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 66 is 4356 and its cube is 287496. 66 is not a perfect square: it lies between 64 and 81. The sum of its digits is 12, so 66 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 66 is even, so it can be shared equally between two people.

Exercise for chapter 66: find two whole numbers whose product is 132, then two whole numbers whose sum is 73. Then write 66 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 66 is even. A hint: divisors usually come in pairs whose product is 66, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 66 that has the same number of divisors, or test whether 67 and 65 share any common factor with 66. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 67. The number 67

This chapter is about the number 67. In Roman numerals it is written LXVII, in binary it is 1000011, and in base eight it is 103. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

67 is a prime number. Its only divisors are 1 and 67. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 67 counters in equal rows, the only way is a single row of 67. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 67 is one of them.

The sum of the divisors of 67 that are smaller than 67 is 1. Because this sum is smaller than 67, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 67 is 4489 and its cube is 300763. 67 is not a perfect square: it lies between 64 and 81. The sum of its digits is 13, so 67 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 67 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 67: find two whole numbers whose product is 134, then two whole numbers whose sum is 74. Then write 67 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 67 is even. A hint: divisors usually come in pairs whose product is 67, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 67 that has the same number of divisors, or test whether 68 and 66 share any common factor with 67. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 68. The number 68

This chapter is about the number 68. In Roman numerals it is written LXVIII, in binary it is 1000100, and in base eight it is 104. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

68 is a composite number. Its divisors are 1, 2, 4, 17, 34, 68, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 68 that are smaller than 68 is 58. Because this sum is smaller than 68, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 68 is 4624 and its cube is 314432. 68 is not a perfect square: it lies between 64 and 81. The sum of its digits is 14, so 68 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 68 is even, so it can be shared equally between two people.

Exercise for chapter 68: find two whole numbers whose product is 136, then two whole numbers whose sum is 75. Then write 68 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 68 is even. A hint: divisors usually come in pairs whose product is 68, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 68 that has the same number of divisors, or test whether 69 and 67 share any common factor with 68. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 69. The number 69

This chapter is about the number 69. In Roman numerals it is written LXIX, in binary it is 1000101, and in base eight it is 105. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

69 is a composite number. Its divisors are 1, 3, 23, 69, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 23. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 69 that are smaller than 69 is 27. Because this sum is smaller than 69, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 69 is 4761 and its cube is 328509. 69 is not a perfect square: it lies between 64 and 81. The sum of its digits is 15, so 69 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 69 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 69: find two whole numbers whose product is 138, then two whole numbers whose sum is 76. Then write 69 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 69 is even. A hint: divisors usually come in pairs whose product is 69, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 69 that has the same number of divisors, or test whether 70 and 68 share any common factor with 69. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 70. The number 70

This chapter is about the number 70. In Roman numerals it is written LXX, in binary it is 1000110, and in base eight it is 106. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

70 is a composite number. Its divisors are 1, 2, 5, 7, 10, 14, 35, 70, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 5 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 70 that are smaller than 70 is 74. Because this sum is larger than 70, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 70 is 4900 and its cube is 343000. 70 is not a perfect square: it lies between 64 and 81. The sum of its digits is 7, so 70 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 70 is even, so it can be shared equally between two people.

Exercise for chapter 70: find two whole numbers whose product is 140, then two whole numbers whose sum is 77. Then write 70 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 70 is even. A hint: divisors usually come in pairs whose product is 70, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 70 that has the same number of divisors, or test whether 71 and 69 share any common factor with 70. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 71. The number 71

This chapter is about the number 71. In Roman numerals it is written LXXI, in binary it is 1000111, and in base eight it is 107. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

71 is a prime number. Its only divisors are 1 and 71. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 71 counters in equal rows, the only way is a single row of 71. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 71 is one of them.

The sum of the divisors of 71 that are smaller than 71 is 1. Because this sum is smaller than 71, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 71 is 5041 and its cube is 357911. 71 is not a perfect square: it lies between 64 and 81. The sum of its digits is 8, so 71 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 71 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 71: find two whole numbers whose product is 142, then two whole numbers whose sum is 78. Then write 71 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 71 is even. A hint: divisors usually come in pairs whose product is 71, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 71 that has the same number of divisors, or test whether 72 and 70 share any common factor with 71. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 72. The number 72

This chapter is about the number 72. In Roman numerals it is written LXXII, in binary it is 1001000, and in base eight it is 110. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

72 is a composite number. Its divisors are 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 72 that are smaller than 72 is 123. Because this sum is larger than 72, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 72 is 5184 and its cube is 373248. 72 is not a perfect square: it lies between 64 and 81. The sum of its digits is 9, so 72 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 72 is even, so it can be shared equally between two people.

Exercise for chapter 72: find two whole numbers whose product is 144, then two whole numbers whose sum is 79. Then write 72 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 72 is even. A hint: divisors usually come in pairs whose product is 72, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 72 that has the same number of divisors, or test whether 73 and 71 share any common factor with 72. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 73. The number 73

This chapter is about the number 73. In Roman numerals it is written LXXIII, in binary it is 1001001, and in base eight it is 111. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

73 is a prime number. Its only divisors are 1 and 73. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 73 counters in equal rows, the only way is a single row of 73. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 73 is one of them.

The sum of the divisors of 73 that are smaller than 73 is 1. Because this sum is smaller than 73, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 73 is 5329 and its cube is 389017. 73 is not a perfect square: it lies between 64 and 81. The sum of its digits is 10, so 73 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 73 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 73: find two whole numbers whose product is 146, then two whole numbers whose sum is 80. Then write 73 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 73 is even. A hint: divisors usually come in pairs whose product is 73, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 73 that has the same number of divisors, or test whether 74 and 72 share any common factor with 73. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 74. The number 74

This chapter is about the number 74. In Roman numerals it is written LXXIV, in binary it is 1001010, and in base eight it is 112. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

74 is a composite number. Its divisors are 1, 2, 37, 74, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 37. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 74 that are smaller than 74 is 40. Because this sum is smaller than 74, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 74 is 5476 and its cube is 405224. 74 is not a perfect square: it lies between 64 and 81. The sum of its digits is 11, so 74 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 74 is even, so it can be shared equally between two people.

Exercise for chapter 74: find two whole numbers whose product is 148, then two whole numbers whose sum is 81. Then write 74 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 74 is even. A hint: divisors usually come in pairs whose product is 74, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 74 that has the same number of divisors, or test whether 75 and 73 share any common factor with 74. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 75. The number 75

This chapter is about the number 75. In Roman numerals it is written LXXV, in binary it is 1001011, and in base eight it is 113. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

75 is a composite number. Its divisors are 1, 3, 5, 15, 25, 75, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 5 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 75 that are smaller than 75 is 49. Because this sum is smaller than 75, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 75 is 5625 and its cube is 421875. 75 is not a perfect square: it lies between 64 and 81. The sum of its digits is 12, so 75 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 75 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 75: find two whole numbers whose product is 150, then two whole numbers whose sum is 82. Then write 75 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 75 is even. A hint: divisors usually come in pairs whose product is 75, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 75 that has the same number of divisors, or test whether 76 and 74 share any common factor with 75. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 76. The number 76

This chapter is about the number 76. In Roman numerals it is written LXXVI, in binary it is 1001100, and in base eight it is 114. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

76 is a composite number. Its divisors are 1, 2, 4, 19, 38, 76, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 76 that are smaller than 76 is 64. Because this sum is smaller than 76, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 76 is 5776 and its cube is 438976. 76 is not a perfect square: it lies between 64 and 81. The sum of its digits is 13, so 76 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 76 is even, so it can be shared equally between two people.

Exercise for chapter 76: find two whole numbers whose product is 152, then two whole numbers whose sum is 83. Then write 76 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 76 is even. A hint: divisors usually come in pairs whose product is 76, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 76 that has the same number of divisors, or test whether 77 and 75 share any common factor with 76. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 77. The number 77

This chapter is about the number 77. In Roman numerals it is written LXXVII, in binary it is 1001101, and in base eight it is 115. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

77 is a composite number. Its divisors are 1, 7, 11, 77, so it has 4 divisors in total. As a product of prime numbers it is written 7 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 77 that are smaller than 77 is 19. Because this sum is smaller than 77, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 77 is 5929 and its cube is 456533. 77 is not a perfect square: it lies between 64 and 81. The sum of its digits is 14, so 77 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 77 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 77: find two whole numbers whose product is 154, then two whole numbers whose sum is 84. Then write 77 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 77 is even. A hint: divisors usually come in pairs whose product is 77, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 77 that has the same number of divisors, or test whether 78 and 76 share any common factor with 77. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 78. The number 78

This chapter is about the number 78. In Roman numerals it is written LXXVIII, in binary it is 1001110, and in base eight it is 116. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

78 is a composite number. Its divisors are 1, 2, 3, 6, 13, 26, 39, 78, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 78 that are smaller than 78 is 90. Because this sum is larger than 78, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 78 is 6084 and its cube is 474552. 78 is not a perfect square: it lies between 64 and 81. The sum of its digits is 15, so 78 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 78 is even, so it can be shared equally between two people.

Exercise for chapter 78: find two whole numbers whose product is 156, then two whole numbers whose sum is 85. Then write 78 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 78 is even. A hint: divisors usually come in pairs whose product is 78, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 78 that has the same number of divisors, or test whether 79 and 77 share any common factor with 78. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 79. The number 79

This chapter is about the number 79. In Roman numerals it is written LXXIX, in binary it is 1001111, and in base eight it is 117. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

79 is a prime number. Its only divisors are 1 and 79. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 79 counters in equal rows, the only way is a single row of 79. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 79 is one of them.

The sum of the divisors of 79 that are smaller than 79 is 1. Because this sum is smaller than 79, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 79 is 6241 and its cube is 493039. 79 is not a perfect square: it lies between 64 and 81. The sum of its digits is 16, so 79 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 79 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 79: find two whole numbers whose product is 158, then two whole numbers whose sum is 86. Then write 79 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 79 is even. A hint: divisors usually come in pairs whose product is 79, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 79 that has the same number of divisors, or test whether 80 and 78 share any common factor with 79. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 80. The number 80

This chapter is about the number 80. In Roman numerals it is written LXXX, in binary it is 1010000, and in base eight it is 120. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

80 is a composite number. Its divisors are 1, 2, 4, 5, 8, 10, 16, 20, 40, 80, so it has 10 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 80 that are smaller than 80 is 106. Because this sum is larger than 80, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 80 is 6400 and its cube is 512000. 80 is not a perfect square: it lies between 64 and 81. The sum of its digits is 8, so 80 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 80 is even, so it can be shared equally between two people.

Exercise for chapter 80: find two whole numbers whose product is 160, then two whole numbers whose sum is 87. Then write 80 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 80 is even. A hint: divisors usually come in pairs whose product is 80, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 80 that has the same number of divisors, or test whether 81 and 79 share any common factor with 80. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 81. The number 81

This chapter is about the number 81. In Roman numerals it is written LXXXI, in binary it is 1010001, and in base eight it is 121. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

81 is a composite number. Its divisors are 1, 3, 9, 27, 81, so it has 5 divisors in total. As a product of prime numbers it is written 3 × 3 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 81 that are smaller than 81 is 40. Because this sum is smaller than 81, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 81 is 6561 and its cube is 531441. 81 is itself a perfect square, because 9 × 9 = 81. The sum of its digits is 9, so 81 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 81 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 81: find two whole numbers whose product is 162, then two whole numbers whose sum is 88. Then write 81 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 81 is odd. A hint: divisors usually come in pairs whose product is 81, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 81 that has the same number of divisors, or test whether 82 and 80 share any common factor with 81. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 82. The number 82

This chapter is about the number 82. In Roman numerals it is written LXXXII, in binary it is 1010010, and in base eight it is 122. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

82 is a composite number. Its divisors are 1, 2, 41, 82, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 41. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 82 that are smaller than 82 is 44. Because this sum is smaller than 82, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 82 is 6724 and its cube is 551368. 82 is not a perfect square: it lies between 81 and 100. The sum of its digits is 10, so 82 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 82 is even, so it can be shared equally between two people.

Exercise for chapter 82: find two whole numbers whose product is 164, then two whole numbers whose sum is 89. Then write 82 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 82 is even. A hint: divisors usually come in pairs whose product is 82, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 82 that has the same number of divisors, or test whether 83 and 81 share any common factor with 82. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 83. The number 83

This chapter is about the number 83. In Roman numerals it is written LXXXIII, in binary it is 1010011, and in base eight it is 123. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

83 is a prime number. Its only divisors are 1 and 83. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 83 counters in equal rows, the only way is a single row of 83. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 83 is one of them.

The sum of the divisors of 83 that are smaller than 83 is 1. Because this sum is smaller than 83, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 83 is 6889 and its cube is 571787. 83 is not a perfect square: it lies between 81 and 100. The sum of its digits is 11, so 83 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 83 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 83: find two whole numbers whose product is 166, then two whole numbers whose sum is 90. Then write 83 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 83 is even. A hint: divisors usually come in pairs whose product is 83, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 83 that has the same number of divisors, or test whether 84 and 82 share any common factor with 83. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 84. The number 84

This chapter is about the number 84. In Roman numerals it is written LXXXIV, in binary it is 1010100, and in base eight it is 124. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

84 is a composite number. Its divisors are 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 3 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 84 that are smaller than 84 is 140. Because this sum is larger than 84, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 84 is 7056 and its cube is 592704. 84 is not a perfect square: it lies between 81 and 100. The sum of its digits is 12, so 84 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 84 is even, so it can be shared equally between two people.

Exercise for chapter 84: find two whole numbers whose product is 168, then two whole numbers whose sum is 91. Then write 84 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 84 is even. A hint: divisors usually come in pairs whose product is 84, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 84 that has the same number of divisors, or test whether 85 and 83 share any common factor with 84. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 85. The number 85

This chapter is about the number 85. In Roman numerals it is written LXXXV, in binary it is 1010101, and in base eight it is 125. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

85 is a composite number. Its divisors are 1, 5, 17, 85, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 85 that are smaller than 85 is 23. Because this sum is smaller than 85, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 85 is 7225 and its cube is 614125. 85 is not a perfect square: it lies between 81 and 100. The sum of its digits is 13, so 85 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 85 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 85: find two whole numbers whose product is 170, then two whole numbers whose sum is 92. Then write 85 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 85 is even. A hint: divisors usually come in pairs whose product is 85, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 85 that has the same number of divisors, or test whether 86 and 84 share any common factor with 85. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 86. The number 86

This chapter is about the number 86. In Roman numerals it is written LXXXVI, in binary it is 1010110, and in base eight it is 126. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

86 is a composite number. Its divisors are 1, 2, 43, 86, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 43. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 86 that are smaller than 86 is 46. Because this sum is smaller than 86, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 86 is 7396 and its cube is 636056. 86 is not a perfect square: it lies between 81 and 100. The sum of its digits is 14, so 86 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 86 is even, so it can be shared equally between two people.

Exercise for chapter 86: find two whole numbers whose product is 172, then two whole numbers whose sum is 93. Then write 86 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 86 is even. A hint: divisors usually come in pairs whose product is 86, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 86 that has the same number of divisors, or test whether 87 and 85 share any common factor with 86. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 87. The number 87

This chapter is about the number 87. In Roman numerals it is written LXXXVII, in binary it is 1010111, and in base eight it is 127. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

87 is a composite number. Its divisors are 1, 3, 29, 87, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 29. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 87 that are smaller than 87 is 33. Because this sum is smaller than 87, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 87 is 7569 and its cube is 658503. 87 is not a perfect square: it lies between 81 and 100. The sum of its digits is 15, so 87 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 87 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 87: find two whole numbers whose product is 174, then two whole numbers whose sum is 94. Then write 87 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 87 is even. A hint: divisors usually come in pairs whose product is 87, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 87 that has the same number of divisors, or test whether 88 and 86 share any common factor with 87. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 88. The number 88

This chapter is about the number 88. In Roman numerals it is written LXXXVIII, in binary it is 1011000, and in base eight it is 130. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

88 is a composite number. Its divisors are 1, 2, 4, 8, 11, 22, 44, 88, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 88 that are smaller than 88 is 92. Because this sum is larger than 88, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 88 is 7744 and its cube is 681472. 88 is not a perfect square: it lies between 81 and 100. The sum of its digits is 16, so 88 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 88 is even, so it can be shared equally between two people.

Exercise for chapter 88: find two whole numbers whose product is 176, then two whole numbers whose sum is 95. Then write 88 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 88 is even. A hint: divisors usually come in pairs whose product is 88, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 88 that has the same number of divisors, or test whether 89 and 87 share any common factor with 88. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 89. The number 89

This chapter is about the number 89. In Roman numerals it is written LXXXIX, in binary it is 1011001, and in base eight it is 131. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

89 is a prime number. Its only divisors are 1 and 89. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 89 counters in equal rows, the only way is a single row of 89. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 89 is one of them.

The sum of the divisors of 89 that are smaller than 89 is 1. Because this sum is smaller than 89, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 89 is 7921 and its cube is 704969. 89 is not a perfect square: it lies between 81 and 100. The sum of its digits is 17, so 89 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 89 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 89: find two whole numbers whose product is 178, then two whole numbers whose sum is 96. Then write 89 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 89 is even. A hint: divisors usually come in pairs whose product is 89, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 89 that has the same number of divisors, or test whether 90 and 88 share any common factor with 89. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 90. The number 90

This chapter is about the number 90. In Roman numerals it is written XC, in binary it is 1011010, and in base eight it is 132. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

90 is a composite number. Its divisors are 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 3 × 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 90 that are smaller than 90 is 144. Because this sum is larger than 90, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 90 is 8100 and its cube is 729000. 90 is not a perfect square: it lies between 81 and 100. The sum of its digits is 9, so 90 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 90 is even, so it can be shared equally between two people.

Exercise for chapter 90: find two whole numbers whose product is 180, then two whole numbers whose sum is 97. Then write 90 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 90 is even. A hint: divisors usually come in pairs whose product is 90, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 90 that has the same number of divisors, or test whether 91 and 89 share any common factor with 90. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 91. The number 91

This chapter is about the number 91. In Roman numerals it is written XCI, in binary it is 1011011, and in base eight it is 133. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

91 is a composite number. Its divisors are 1, 7, 13, 91, so it has 4 divisors in total. As a product of prime numbers it is written 7 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 91 that are smaller than 91 is 21. Because this sum is smaller than 91, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 91 is 8281 and its cube is 753571. 91 is not a perfect square: it lies between 81 and 100. The sum of its digits is 10, so 91 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 91 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 91: find two whole numbers whose product is 182, then two whole numbers whose sum is 98. Then write 91 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 91 is even. A hint: divisors usually come in pairs whose product is 91, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 91 that has the same number of divisors, or test whether 92 and 90 share any common factor with 91. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 92. The number 92

This chapter is about the number 92. In Roman numerals it is written XCII, in binary it is 1011100, and in base eight it is 134. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

92 is a composite number. Its divisors are 1, 2, 4, 23, 46, 92, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 23. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 92 that are smaller than 92 is 76. Because this sum is smaller than 92, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 92 is 8464 and its cube is 778688. 92 is not a perfect square: it lies between 81 and 100. The sum of its digits is 11, so 92 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 92 is even, so it can be shared equally between two people.

Exercise for chapter 92: find two whole numbers whose product is 184, then two whole numbers whose sum is 99. Then write 92 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 92 is even. A hint: divisors usually come in pairs whose product is 92, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 92 that has the same number of divisors, or test whether 93 and 91 share any common factor with 92. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 93. The number 93

This chapter is about the number 93. In Roman numerals it is written XCIII, in binary it is 1011101, and in base eight it is 135. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

93 is a composite number. Its divisors are 1, 3, 31, 93, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 31. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 93 that are smaller than 93 is 35. Because this sum is smaller than 93, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 93 is 8649 and its cube is 804357. 93 is not a perfect square: it lies between 81 and 100. The sum of its digits is 12, so 93 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 93 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 93: find two whole numbers whose product is 186, then two whole numbers whose sum is 100. Then write 93 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 93 is even. A hint: divisors usually come in pairs whose product is 93, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 93 that has the same number of divisors, or test whether 94 and 92 share any common factor with 93. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 94. The number 94

This chapter is about the number 94. In Roman numerals it is written XCIV, in binary it is 1011110, and in base eight it is 136. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

94 is a composite number. Its divisors are 1, 2, 47, 94, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 47. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 94 that are smaller than 94 is 50. Because this sum is smaller than 94, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 94 is 8836 and its cube is 830584. 94 is not a perfect square: it lies between 81 and 100. The sum of its digits is 13, so 94 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 94 is even, so it can be shared equally between two people.

Exercise for chapter 94: find two whole numbers whose product is 188, then two whole numbers whose sum is 101. Then write 94 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 94 is even. A hint: divisors usually come in pairs whose product is 94, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 94 that has the same number of divisors, or test whether 95 and 93 share any common factor with 94. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 95. The number 95

This chapter is about the number 95. In Roman numerals it is written XCV, in binary it is 1011111, and in base eight it is 137. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

95 is a composite number. Its divisors are 1, 5, 19, 95, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 95 that are smaller than 95 is 25. Because this sum is smaller than 95, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 95 is 9025 and its cube is 857375. 95 is not a perfect square: it lies between 81 and 100. The sum of its digits is 14, so 95 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 95 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 95: find two whole numbers whose product is 190, then two whole numbers whose sum is 102. Then write 95 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 95 is even. A hint: divisors usually come in pairs whose product is 95, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 95 that has the same number of divisors, or test whether 96 and 94 share any common factor with 95. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 96. The number 96

This chapter is about the number 96. In Roman numerals it is written XCVI, in binary it is 1100000, and in base eight it is 140. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

96 is a composite number. Its divisors are 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 2 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 96 that are smaller than 96 is 156. Because this sum is larger than 96, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 96 is 9216 and its cube is 884736. 96 is not a perfect square: it lies between 81 and 100. The sum of its digits is 15, so 96 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 96 is even, so it can be shared equally between two people.

Exercise for chapter 96: find two whole numbers whose product is 192, then two whole numbers whose sum is 103. Then write 96 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 96 is even. A hint: divisors usually come in pairs whose product is 96, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 96 that has the same number of divisors, or test whether 97 and 95 share any common factor with 96. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 97. The number 97

This chapter is about the number 97. In Roman numerals it is written XCVII, in binary it is 1100001, and in base eight it is 141. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

97 is a prime number. Its only divisors are 1 and 97. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 97 counters in equal rows, the only way is a single row of 97. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 97 is one of them.

The sum of the divisors of 97 that are smaller than 97 is 1. Because this sum is smaller than 97, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 97 is 9409 and its cube is 912673. 97 is not a perfect square: it lies between 81 and 100. The sum of its digits is 16, so 97 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 97 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 97: find two whole numbers whose product is 194, then two whole numbers whose sum is 104. Then write 97 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 97 is even. A hint: divisors usually come in pairs whose product is 97, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 97 that has the same number of divisors, or test whether 98 and 96 share any common factor with 97. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 98. The number 98

This chapter is about the number 98. In Roman numerals it is written XCVIII, in binary it is 1100010, and in base eight it is 142. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

98 is a composite number. Its divisors are 1, 2, 7, 14, 49, 98, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 7 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 98 that are smaller than 98 is 73. Because this sum is smaller than 98, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 98 is 9604 and its cube is 941192. 98 is not a perfect square: it lies between 81 and 100. The sum of its digits is 17, so 98 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 98 is even, so it can be shared equally between two people.

Exercise for chapter 98: find two whole numbers whose product is 196, then two whole numbers whose sum is 105. Then write 98 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 98 is even. A hint: divisors usually come in pairs whose product is 98, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 98 that has the same number of divisors, or test whether 99 and 97 share any common factor with 98. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 99. The number 99

This chapter is about the number 99. In Roman numerals it is written XCIX, in binary it is 1100011, and in base eight it is 143. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

99 is a composite number. Its divisors are 1, 3, 9, 11, 33, 99, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 3 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 99 that are smaller than 99 is 57. Because this sum is smaller than 99, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 99 is 9801 and its cube is 970299. 99 is not a perfect square: it lies between 81 and 100. The sum of its digits is 18, so 99 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 99 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 99: find two whole numbers whose product is 198, then two whole numbers whose sum is 106. Then write 99 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 99 is even. A hint: divisors usually come in pairs whose product is 99, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 99 that has the same number of divisors, or test whether 100 and 98 share any common factor with 99. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 100. The number 100

This chapter is about the number 100. In Roman numerals it is written C, in binary it is 1100100, and in base eight it is 144. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

100 is a composite number. Its divisors are 1, 2, 4, 5, 10, 20, 25, 50, 100, so it has 9 divisors in total. As a product of prime numbers it is written 2 × 2 × 5 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 100 that are smaller than 100 is 117. Because this sum is larger than 100, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 100 is 10000 and its cube is 1000000. 100 is itself a perfect square, because 10 × 10 = 100. The sum of its digits is 1, so 100 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 100 is even, so it can be shared equally between two people.

Exercise for chapter 100: find two whole numbers whose product is 200, then two whole numbers whose sum is 107. Then write 100 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 100 is odd. A hint: divisors usually come in pairs whose product is 100, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 100 that has the same number of divisors, or test whether 101 and 99 share any common factor with 100. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 101. The number 101

This chapter is about the number 101. In Roman numerals it is written CI, in binary it is 1100101, and in base eight it is 145. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

101 is a prime number. Its only divisors are 1 and 101. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 101 counters in equal rows, the only way is a single row of 101. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 101 is one of them.

The sum of the divisors of 101 that are smaller than 101 is 1. Because this sum is smaller than 101, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 101 is 10201 and its cube is 1030301. 101 is not a perfect square: it lies between 100 and 121. The sum of its digits is 2, so 101 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 101 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 101: find two whole numbers whose product is 202, then two whole numbers whose sum is 108. Then write 101 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 101 is even. A hint: divisors usually come in pairs whose product is 101, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 101 that has the same number of divisors, or test whether 102 and 100 share any common factor with 101. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 102. The number 102

This chapter is about the number 102. In Roman numerals it is written CII, in binary it is 1100110, and in base eight it is 146. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

102 is a composite number. Its divisors are 1, 2, 3, 6, 17, 34, 51, 102, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 102 that are smaller than 102 is 114. Because this sum is larger than 102, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 102 is 10404 and its cube is 1061208. 102 is not a perfect square: it lies between 100 and 121. The sum of its digits is 3, so 102 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 102 is even, so it can be shared equally between two people.

Exercise for chapter 102: find two whole numbers whose product is 204, then two whole numbers whose sum is 109. Then write 102 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 102 is even. A hint: divisors usually come in pairs whose product is 102, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 102 that has the same number of divisors, or test whether 103 and 101 share any common factor with 102. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 103. The number 103

This chapter is about the number 103. In Roman numerals it is written CIII, in binary it is 1100111, and in base eight it is 147. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

103 is a prime number. Its only divisors are 1 and 103. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 103 counters in equal rows, the only way is a single row of 103. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 103 is one of them.

The sum of the divisors of 103 that are smaller than 103 is 1. Because this sum is smaller than 103, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 103 is 10609 and its cube is 1092727. 103 is not a perfect square: it lies between 100 and 121. The sum of its digits is 4, so 103 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 103 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 103: find two whole numbers whose product is 206, then two whole numbers whose sum is 110. Then write 103 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 103 is even. A hint: divisors usually come in pairs whose product is 103, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 103 that has the same number of divisors, or test whether 104 and 102 share any common factor with 103. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 104. The number 104

This chapter is about the number 104. In Roman numerals it is written CIV, in binary it is 1101000, and in base eight it is 150. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

104 is a composite number. Its divisors are 1, 2, 4, 8, 13, 26, 52, 104, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 104 that are smaller than 104 is 106. Because this sum is larger than 104, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 104 is 10816 and its cube is 1124864. 104 is not a perfect square: it lies between 100 and 121. The sum of its digits is 5, so 104 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 104 is even, so it can be shared equally between two people.

Exercise for chapter 104: find two whole numbers whose product is 208, then two whole numbers whose sum is 111. Then write 104 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 104 is even. A hint: divisors usually come in pairs whose product is 104, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 104 that has the same number of divisors, or test whether 105 and 103 share any common factor with 104. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 105. The number 105

This chapter is about the number 105. In Roman numerals it is written CV, in binary it is 1101001, and in base eight it is 151. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

105 is a composite number. Its divisors are 1, 3, 5, 7, 15, 21, 35, 105, so it has 8 divisors in total. As a product of prime numbers it is written 3 × 5 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 105 that are smaller than 105 is 87. Because this sum is smaller than 105, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 105 is 11025 and its cube is 1157625. 105 is not a perfect square: it lies between 100 and 121. The sum of its digits is 6, so 105 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 105 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 105: find two whole numbers whose product is 210, then two whole numbers whose sum is 112. Then write 105 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 105 is even. A hint: divisors usually come in pairs whose product is 105, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 105 that has the same number of divisors, or test whether 106 and 104 share any common factor with 105. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 106. The number 106

This chapter is about the number 106. In Roman numerals it is written CVI, in binary it is 1101010, and in base eight it is 152. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

106 is a composite number. Its divisors are 1, 2, 53, 106, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 53. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 106 that are smaller than 106 is 56. Because this sum is smaller than 106, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 106 is 11236 and its cube is 1191016. 106 is not a perfect square: it lies between 100 and 121. The sum of its digits is 7, so 106 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 106 is even, so it can be shared equally between two people.

Exercise for chapter 106: find two whole numbers whose product is 212, then two whole numbers whose sum is 113. Then write 106 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 106 is even. A hint: divisors usually come in pairs whose product is 106, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 106 that has the same number of divisors, or test whether 107 and 105 share any common factor with 106. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 107. The number 107

This chapter is about the number 107. In Roman numerals it is written CVII, in binary it is 1101011, and in base eight it is 153. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

107 is a prime number. Its only divisors are 1 and 107. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 107 counters in equal rows, the only way is a single row of 107. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 107 is one of them.

The sum of the divisors of 107 that are smaller than 107 is 1. Because this sum is smaller than 107, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 107 is 11449 and its cube is 1225043. 107 is not a perfect square: it lies between 100 and 121. The sum of its digits is 8, so 107 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 107 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 107: find two whole numbers whose product is 214, then two whole numbers whose sum is 114. Then write 107 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 107 is even. A hint: divisors usually come in pairs whose product is 107, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 107 that has the same number of divisors, or test whether 108 and 106 share any common factor with 107. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 108. The number 108

This chapter is about the number 108. In Roman numerals it is written CVIII, in binary it is 1101100, and in base eight it is 154. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

108 is a composite number. Its divisors are 1, 2, 3, 4, 6, 9, 12, 18, 27, 36, 54, 108, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 3 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 108 that are smaller than 108 is 172. Because this sum is larger than 108, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 108 is 11664 and its cube is 1259712. 108 is not a perfect square: it lies between 100 and 121. The sum of its digits is 9, so 108 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 108 is even, so it can be shared equally between two people.

Exercise for chapter 108: find two whole numbers whose product is 216, then two whole numbers whose sum is 115. Then write 108 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 108 is even. A hint: divisors usually come in pairs whose product is 108, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 108 that has the same number of divisors, or test whether 109 and 107 share any common factor with 108. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 109. The number 109

This chapter is about the number 109. In Roman numerals it is written CIX, in binary it is 1101101, and in base eight it is 155. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

109 is a prime number. Its only divisors are 1 and 109. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 109 counters in equal rows, the only way is a single row of 109. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 109 is one of them.

The sum of the divisors of 109 that are smaller than 109 is 1. Because this sum is smaller than 109, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 109 is 11881 and its cube is 1295029. 109 is not a perfect square: it lies between 100 and 121. The sum of its digits is 10, so 109 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 109 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 109: find two whole numbers whose product is 218, then two whole numbers whose sum is 116. Then write 109 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 109 is even. A hint: divisors usually come in pairs whose product is 109, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 109 that has the same number of divisors, or test whether 110 and 108 share any common factor with 109. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 110. The number 110

This chapter is about the number 110. In Roman numerals it is written CX, in binary it is 1101110, and in base eight it is 156. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

110 is a composite number. Its divisors are 1, 2, 5, 10, 11, 22, 55, 110, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 5 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 110 that are smaller than 110 is 106. Because this sum is smaller than 110, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 110 is 12100 and its cube is 1331000. 110 is not a perfect square: it lies between 100 and 121. The sum of its digits is 2, so 110 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 110 is even, so it can be shared equally between two people.

Exercise for chapter 110: find two whole numbers whose product is 220, then two whole numbers whose sum is 117. Then write 110 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 110 is even. A hint: divisors usually come in pairs whose product is 110, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 110 that has the same number of divisors, or test whether 111 and 109 share any common factor with 110. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 111. The number 111

This chapter is about the number 111. In Roman numerals it is written CXI, in binary it is 1101111, and in base eight it is 157. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

111 is a composite number. Its divisors are 1, 3, 37, 111, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 37. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 111 that are smaller than 111 is 41. Because this sum is smaller than 111, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 111 is 12321 and its cube is 1367631. 111 is not a perfect square: it lies between 100 and 121. The sum of its digits is 3, so 111 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 111 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 111: find two whole numbers whose product is 222, then two whole numbers whose sum is 118. Then write 111 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 111 is even. A hint: divisors usually come in pairs whose product is 111, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 111 that has the same number of divisors, or test whether 112 and 110 share any common factor with 111. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 112. The number 112

This chapter is about the number 112. In Roman numerals it is written CXII, in binary it is 1110000, and in base eight it is 160. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

112 is a composite number. Its divisors are 1, 2, 4, 7, 8, 14, 16, 28, 56, 112, so it has 10 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 112 that are smaller than 112 is 136. Because this sum is larger than 112, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 112 is 12544 and its cube is 1404928. 112 is not a perfect square: it lies between 100 and 121. The sum of its digits is 4, so 112 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 112 is even, so it can be shared equally between two people.

Exercise for chapter 112: find two whole numbers whose product is 224, then two whole numbers whose sum is 119. Then write 112 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 112 is even. A hint: divisors usually come in pairs whose product is 112, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 112 that has the same number of divisors, or test whether 113 and 111 share any common factor with 112. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 113. The number 113

This chapter is about the number 113. In Roman numerals it is written CXIII, in binary it is 1110001, and in base eight it is 161. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

113 is a prime number. Its only divisors are 1 and 113. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 113 counters in equal rows, the only way is a single row of 113. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 113 is one of them.

The sum of the divisors of 113 that are smaller than 113 is 1. Because this sum is smaller than 113, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 113 is 12769 and its cube is 1442897. 113 is not a perfect square: it lies between 100 and 121. The sum of its digits is 5, so 113 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 113 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 113: find two whole numbers whose product is 226, then two whole numbers whose sum is 120. Then write 113 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 113 is even. A hint: divisors usually come in pairs whose product is 113, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 113 that has the same number of divisors, or test whether 114 and 112 share any common factor with 113. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 114. The number 114

This chapter is about the number 114. In Roman numerals it is written CXIV, in binary it is 1110010, and in base eight it is 162. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

114 is a composite number. Its divisors are 1, 2, 3, 6, 19, 38, 57, 114, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 114 that are smaller than 114 is 126. Because this sum is larger than 114, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 114 is 12996 and its cube is 1481544. 114 is not a perfect square: it lies between 100 and 121. The sum of its digits is 6, so 114 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 114 is even, so it can be shared equally between two people.

Exercise for chapter 114: find two whole numbers whose product is 228, then two whole numbers whose sum is 121. Then write 114 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 114 is even. A hint: divisors usually come in pairs whose product is 114, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 114 that has the same number of divisors, or test whether 115 and 113 share any common factor with 114. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 115. The number 115

This chapter is about the number 115. In Roman numerals it is written CXV, in binary it is 1110011, and in base eight it is 163. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

115 is a composite number. Its divisors are 1, 5, 23, 115, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 23. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 115 that are smaller than 115 is 29. Because this sum is smaller than 115, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 115 is 13225 and its cube is 1520875. 115 is not a perfect square: it lies between 100 and 121. The sum of its digits is 7, so 115 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 115 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 115: find two whole numbers whose product is 230, then two whole numbers whose sum is 122. Then write 115 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 115 is even. A hint: divisors usually come in pairs whose product is 115, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 115 that has the same number of divisors, or test whether 116 and 114 share any common factor with 115. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 116. The number 116

This chapter is about the number 116. In Roman numerals it is written CXVI, in binary it is 1110100, and in base eight it is 164. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

116 is a composite number. Its divisors are 1, 2, 4, 29, 58, 116, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 29. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 116 that are smaller than 116 is 94. Because this sum is smaller than 116, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 116 is 13456 and its cube is 1560896. 116 is not a perfect square: it lies between 100 and 121. The sum of its digits is 8, so 116 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 116 is even, so it can be shared equally between two people.

Exercise for chapter 116: find two whole numbers whose product is 232, then two whole numbers whose sum is 123. Then write 116 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 116 is even. A hint: divisors usually come in pairs whose product is 116, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 116 that has the same number of divisors, or test whether 117 and 115 share any common factor with 116. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 117. The number 117

This chapter is about the number 117. In Roman numerals it is written CXVII, in binary it is 1110101, and in base eight it is 165. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

117 is a composite number. Its divisors are 1, 3, 9, 13, 39, 117, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 3 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 117 that are smaller than 117 is 65. Because this sum is smaller than 117, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 117 is 13689 and its cube is 1601613. 117 is not a perfect square: it lies between 100 and 121. The sum of its digits is 9, so 117 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 117 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 117: find two whole numbers whose product is 234, then two whole numbers whose sum is 124. Then write 117 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 117 is even. A hint: divisors usually come in pairs whose product is 117, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 117 that has the same number of divisors, or test whether 118 and 116 share any common factor with 117. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 118. The number 118

This chapter is about the number 118. In Roman numerals it is written CXVIII, in binary it is 1110110, and in base eight it is 166. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

118 is a composite number. Its divisors are 1, 2, 59, 118, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 59. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 118 that are smaller than 118 is 62. Because this sum is smaller than 118, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 118 is 13924 and its cube is 1643032. 118 is not a perfect square: it lies between 100 and 121. The sum of its digits is 10, so 118 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 118 is even, so it can be shared equally between two people.

Exercise for chapter 118: find two whole numbers whose product is 236, then two whole numbers whose sum is 125. Then write 118 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 118 is even. A hint: divisors usually come in pairs whose product is 118, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 118 that has the same number of divisors, or test whether 119 and 117 share any common factor with 118. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 119. The number 119

This chapter is about the number 119. In Roman numerals it is written CXIX, in binary it is 1110111, and in base eight it is 167. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

119 is a composite number. Its divisors are 1, 7, 17, 119, so it has 4 divisors in total. As a product of prime numbers it is written 7 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 119 that are smaller than 119 is 25. Because this sum is smaller than 119, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 119 is 14161 and its cube is 1685159. 119 is not a perfect square: it lies between 100 and 121. The sum of its digits is 11, so 119 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 119 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 119: find two whole numbers whose product is 238, then two whole numbers whose sum is 126. Then write 119 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 119 is even. A hint: divisors usually come in pairs whose product is 119, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 119 that has the same number of divisors, or test whether 120 and 118 share any common factor with 119. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 120. The number 120

This chapter is about the number 120. In Roman numerals it is written CXX, in binary it is 1111000, and in base eight it is 170. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

120 is a composite number. Its divisors are 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120, so it has 16 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 120 that are smaller than 120 is 240. Because this sum is larger than 120, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 120 is 14400 and its cube is 1728000. 120 is not a perfect square: it lies between 100 and 121. The sum of its digits is 3, so 120 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 120 is even, so it can be shared equally between two people.

Exercise for chapter 120: find two whole numbers whose product is 240, then two whole numbers whose sum is 127. Then write 120 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 120 is even. A hint: divisors usually come in pairs whose product is 120, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 120 that has the same number of divisors, or test whether 121 and 119 share any common factor with 120. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 121. The number 121

This chapter is about the number 121. In Roman numerals it is written CXXI, in binary it is 1111001, and in base eight it is 171. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

121 is a composite number. Its divisors are 1, 11, 121, so it has 3 divisors in total. As a product of prime numbers it is written 11 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 121 that are smaller than 121 is 12. Because this sum is smaller than 121, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 121 is 14641 and its cube is 1771561. 121 is itself a perfect square, because 11 × 11 = 121. The sum of its digits is 4, so 121 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 121 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 121: find two whole numbers whose product is 242, then two whole numbers whose sum is 128. Then write 121 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 121 is odd. A hint: divisors usually come in pairs whose product is 121, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 121 that has the same number of divisors, or test whether 122 and 120 share any common factor with 121. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 122. The number 122

This chapter is about the number 122. In Roman numerals it is written CXXII, in binary it is 1111010, and in base eight it is 172. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

122 is a composite number. Its divisors are 1, 2, 61, 122, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 61. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 122 that are smaller than 122 is 64. Because this sum is smaller than 122, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 122 is 14884 and its cube is 1815848. 122 is not a perfect square: it lies between 121 and 144. The sum of its digits is 5, so 122 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 122 is even, so it can be shared equally between two people.

Exercise for chapter 122: find two whole numbers whose product is 244, then two whole numbers whose sum is 129. Then write 122 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 122 is even. A hint: divisors usually come in pairs whose product is 122, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 122 that has the same number of divisors, or test whether 123 and 121 share any common factor with 122. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 123. The number 123

This chapter is about the number 123. In Roman numerals it is written CXXIII, in binary it is 1111011, and in base eight it is 173. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

123 is a composite number. Its divisors are 1, 3, 41, 123, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 41. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 123 that are smaller than 123 is 45. Because this sum is smaller than 123, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 123 is 15129 and its cube is 1860867. 123 is not a perfect square: it lies between 121 and 144. The sum of its digits is 6, so 123 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 123 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 123: find two whole numbers whose product is 246, then two whole numbers whose sum is 130. Then write 123 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 123 is even. A hint: divisors usually come in pairs whose product is 123, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 123 that has the same number of divisors, or test whether 124 and 122 share any common factor with 123. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 124. The number 124

This chapter is about the number 124. In Roman numerals it is written CXXIV, in binary it is 1111100, and in base eight it is 174. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

124 is a composite number. Its divisors are 1, 2, 4, 31, 62, 124, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 31. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 124 that are smaller than 124 is 100. Because this sum is smaller than 124, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 124 is 15376 and its cube is 1906624. 124 is not a perfect square: it lies between 121 and 144. The sum of its digits is 7, so 124 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 124 is even, so it can be shared equally between two people.

Exercise for chapter 124: find two whole numbers whose product is 248, then two whole numbers whose sum is 131. Then write 124 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 124 is even. A hint: divisors usually come in pairs whose product is 124, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 124 that has the same number of divisors, or test whether 125 and 123 share any common factor with 124. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 125. The number 125

This chapter is about the number 125. In Roman numerals it is written CXXV, in binary it is 1111101, and in base eight it is 175. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

125 is a composite number. Its divisors are 1, 5, 25, 125, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 5 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 125 that are smaller than 125 is 31. Because this sum is smaller than 125, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 125 is 15625 and its cube is 1953125. 125 is not a perfect square: it lies between 121 and 144. The sum of its digits is 8, so 125 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 125 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 125: find two whole numbers whose product is 250, then two whole numbers whose sum is 132. Then write 125 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 125 is even. A hint: divisors usually come in pairs whose product is 125, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 125 that has the same number of divisors, or test whether 126 and 124 share any common factor with 125. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 126. The number 126

This chapter is about the number 126. In Roman numerals it is written CXXVI, in binary it is 1111110, and in base eight it is 176. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

126 is a composite number. Its divisors are 1, 2, 3, 6, 7, 9, 14, 18, 21, 42, 63, 126, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 3 × 3 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 126 that are smaller than 126 is 186. Because this sum is larger than 126, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 126 is 15876 and its cube is 2000376. 126 is not a perfect square: it lies between 121 and 144. The sum of its digits is 9, so 126 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 126 is even, so it can be shared equally between two people.

Exercise for chapter 126: find two whole numbers whose product is 252, then two whole numbers whose sum is 133. Then write 126 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 126 is even. A hint: divisors usually come in pairs whose product is 126, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 126 that has the same number of divisors, or test whether 127 and 125 share any common factor with 126. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 127. The number 127

This chapter is about the number 127. In Roman numerals it is written CXXVII, in binary it is 1111111, and in base eight it is 177. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

127 is a prime number. Its only divisors are 1 and 127. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 127 counters in equal rows, the only way is a single row of 127. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 127 is one of them.

The sum of the divisors of 127 that are smaller than 127 is 1. Because this sum is smaller than 127, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 127 is 16129 and its cube is 2048383. 127 is not a perfect square: it lies between 121 and 144. The sum of its digits is 10, so 127 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 127 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 127: find two whole numbers whose product is 254, then two whole numbers whose sum is 134. Then write 127 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 127 is even. A hint: divisors usually come in pairs whose product is 127, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 127 that has the same number of divisors, or test whether 128 and 126 share any common factor with 127. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 128. The number 128

This chapter is about the number 128. In Roman numerals it is written CXXVIII, in binary it is 10000000, and in base eight it is 200. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

128 is a composite number. Its divisors are 1, 2, 4, 8, 16, 32, 64, 128, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 2 × 2 × 2. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 128 that are smaller than 128 is 127. Because this sum is smaller than 128, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 128 is 16384 and its cube is 2097152. 128 is not a perfect square: it lies between 121 and 144. The sum of its digits is 11, so 128 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 128 is even, so it can be shared equally between two people.

Exercise for chapter 128: find two whole numbers whose product is 256, then two whole numbers whose sum is 135. Then write 128 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 128 is even. A hint: divisors usually come in pairs whose product is 128, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 128 that has the same number of divisors, or test whether 129 and 127 share any common factor with 128. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 129. The number 129

This chapter is about the number 129. In Roman numerals it is written CXXIX, in binary it is 10000001, and in base eight it is 201. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

129 is a composite number. Its divisors are 1, 3, 43, 129, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 43. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 129 that are smaller than 129 is 47. Because this sum is smaller than 129, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 129 is 16641 and its cube is 2146689. 129 is not a perfect square: it lies between 121 and 144. The sum of its digits is 12, so 129 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 129 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 129: find two whole numbers whose product is 258, then two whole numbers whose sum is 136. Then write 129 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 129 is even. A hint: divisors usually come in pairs whose product is 129, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 129 that has the same number of divisors, or test whether 130 and 128 share any common factor with 129. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 130. The number 130

This chapter is about the number 130. In Roman numerals it is written CXXX, in binary it is 10000010, and in base eight it is 202. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

130 is a composite number. Its divisors are 1, 2, 5, 10, 13, 26, 65, 130, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 5 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 130 that are smaller than 130 is 122. Because this sum is smaller than 130, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 130 is 16900 and its cube is 2197000. 130 is not a perfect square: it lies between 121 and 144. The sum of its digits is 4, so 130 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 130 is even, so it can be shared equally between two people.

Exercise for chapter 130: find two whole numbers whose product is 260, then two whole numbers whose sum is 137. Then write 130 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 130 is even. A hint: divisors usually come in pairs whose product is 130, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 130 that has the same number of divisors, or test whether 131 and 129 share any common factor with 130. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 131. The number 131

This chapter is about the number 131. In Roman numerals it is written CXXXI, in binary it is 10000011, and in base eight it is 203. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

131 is a prime number. Its only divisors are 1 and 131. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 131 counters in equal rows, the only way is a single row of 131. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 131 is one of them.

The sum of the divisors of 131 that are smaller than 131 is 1. Because this sum is smaller than 131, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 131 is 17161 and its cube is 2248091. 131 is not a perfect square: it lies between 121 and 144. The sum of its digits is 5, so 131 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 131 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 131: find two whole numbers whose product is 262, then two whole numbers whose sum is 138. Then write 131 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 131 is even. A hint: divisors usually come in pairs whose product is 131, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 131 that has the same number of divisors, or test whether 132 and 130 share any common factor with 131. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 132. The number 132

This chapter is about the number 132. In Roman numerals it is written CXXXII, in binary it is 10000100, and in base eight it is 204. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

132 is a composite number. Its divisors are 1, 2, 3, 4, 6, 11, 12, 22, 33, 44, 66, 132, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 3 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 132 that are smaller than 132 is 204. Because this sum is larger than 132, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 132 is 17424 and its cube is 2299968. 132 is not a perfect square: it lies between 121 and 144. The sum of its digits is 6, so 132 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 132 is even, so it can be shared equally between two people.

Exercise for chapter 132: find two whole numbers whose product is 264, then two whole numbers whose sum is 139. Then write 132 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 132 is even. A hint: divisors usually come in pairs whose product is 132, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 132 that has the same number of divisors, or test whether 133 and 131 share any common factor with 132. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 133. The number 133

This chapter is about the number 133. In Roman numerals it is written CXXXIII, in binary it is 10000101, and in base eight it is 205. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

133 is a composite number. Its divisors are 1, 7, 19, 133, so it has 4 divisors in total. As a product of prime numbers it is written 7 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 133 that are smaller than 133 is 27. Because this sum is smaller than 133, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 133 is 17689 and its cube is 2352637. 133 is not a perfect square: it lies between 121 and 144. The sum of its digits is 7, so 133 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 133 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 133: find two whole numbers whose product is 266, then two whole numbers whose sum is 140. Then write 133 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 133 is even. A hint: divisors usually come in pairs whose product is 133, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 133 that has the same number of divisors, or test whether 134 and 132 share any common factor with 133. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 134. The number 134

This chapter is about the number 134. In Roman numerals it is written CXXXIV, in binary it is 10000110, and in base eight it is 206. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

134 is a composite number. Its divisors are 1, 2, 67, 134, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 67. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 134 that are smaller than 134 is 70. Because this sum is smaller than 134, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 134 is 17956 and its cube is 2406104. 134 is not a perfect square: it lies between 121 and 144. The sum of its digits is 8, so 134 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 134 is even, so it can be shared equally between two people.

Exercise for chapter 134: find two whole numbers whose product is 268, then two whole numbers whose sum is 141. Then write 134 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 134 is even. A hint: divisors usually come in pairs whose product is 134, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 134 that has the same number of divisors, or test whether 135 and 133 share any common factor with 134. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 135. The number 135

This chapter is about the number 135. In Roman numerals it is written CXXXV, in binary it is 10000111, and in base eight it is 207. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

135 is a composite number. Its divisors are 1, 3, 5, 9, 15, 27, 45, 135, so it has 8 divisors in total. As a product of prime numbers it is written 3 × 3 × 3 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 135 that are smaller than 135 is 105. Because this sum is smaller than 135, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 135 is 18225 and its cube is 2460375. 135 is not a perfect square: it lies between 121 and 144. The sum of its digits is 9, so 135 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 135 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 135: find two whole numbers whose product is 270, then two whole numbers whose sum is 142. Then write 135 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 135 is even. A hint: divisors usually come in pairs whose product is 135, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 135 that has the same number of divisors, or test whether 136 and 134 share any common factor with 135. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 136. The number 136

This chapter is about the number 136. In Roman numerals it is written CXXXVI, in binary it is 10001000, and in base eight it is 210. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

136 is a composite number. Its divisors are 1, 2, 4, 8, 17, 34, 68, 136, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 136 that are smaller than 136 is 134. Because this sum is smaller than 136, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 136 is 18496 and its cube is 2515456. 136 is not a perfect square: it lies between 121 and 144. The sum of its digits is 10, so 136 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 136 is even, so it can be shared equally between two people.

Exercise for chapter 136: find two whole numbers whose product is 272, then two whole numbers whose sum is 143. Then write 136 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 136 is even. A hint: divisors usually come in pairs whose product is 136, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 136 that has the same number of divisors, or test whether 137 and 135 share any common factor with 136. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 137. The number 137

This chapter is about the number 137. In Roman numerals it is written CXXXVII, in binary it is 10001001, and in base eight it is 211. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

137 is a prime number. Its only divisors are 1 and 137. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 137 counters in equal rows, the only way is a single row of 137. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 137 is one of them.

The sum of the divisors of 137 that are smaller than 137 is 1. Because this sum is smaller than 137, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 137 is 18769 and its cube is 2571353. 137 is not a perfect square: it lies between 121 and 144. The sum of its digits is 11, so 137 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 137 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 137: find two whole numbers whose product is 274, then two whole numbers whose sum is 144. Then write 137 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 137 is even. A hint: divisors usually come in pairs whose product is 137, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 137 that has the same number of divisors, or test whether 138 and 136 share any common factor with 137. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 138. The number 138

This chapter is about the number 138. In Roman numerals it is written CXXXVIII, in binary it is 10001010, and in base eight it is 212. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

138 is a composite number. Its divisors are 1, 2, 3, 6, 23, 46, 69, 138, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 3 × 23. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 138 that are smaller than 138 is 150. Because this sum is larger than 138, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 138 is 19044 and its cube is 2628072. 138 is not a perfect square: it lies between 121 and 144. The sum of its digits is 12, so 138 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 138 is even, so it can be shared equally between two people.

Exercise for chapter 138: find two whole numbers whose product is 276, then two whole numbers whose sum is 145. Then write 138 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 138 is even. A hint: divisors usually come in pairs whose product is 138, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 138 that has the same number of divisors, or test whether 139 and 137 share any common factor with 138. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 139. The number 139

This chapter is about the number 139. In Roman numerals it is written CXXXIX, in binary it is 10001011, and in base eight it is 213. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

139 is a prime number. Its only divisors are 1 and 139. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 139 counters in equal rows, the only way is a single row of 139. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 139 is one of them.

The sum of the divisors of 139 that are smaller than 139 is 1. Because this sum is smaller than 139, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 139 is 19321 and its cube is 2685619. 139 is not a perfect square: it lies between 121 and 144. The sum of its digits is 13, so 139 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 139 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 139: find two whole numbers whose product is 278, then two whole numbers whose sum is 146. Then write 139 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 139 is even. A hint: divisors usually come in pairs whose product is 139, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 139 that has the same number of divisors, or test whether 140 and 138 share any common factor with 139. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 140. The number 140

This chapter is about the number 140. In Roman numerals it is written CXL, in binary it is 10001100, and in base eight it is 214. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

140 is a composite number. Its divisors are 1, 2, 4, 5, 7, 10, 14, 20, 28, 35, 70, 140, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 5 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 140 that are smaller than 140 is 196. Because this sum is larger than 140, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 140 is 19600 and its cube is 2744000. 140 is not a perfect square: it lies between 121 and 144. The sum of its digits is 5, so 140 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 140 is even, so it can be shared equally between two people.

Exercise for chapter 140: find two whole numbers whose product is 280, then two whole numbers whose sum is 147. Then write 140 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 140 is even. A hint: divisors usually come in pairs whose product is 140, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 140 that has the same number of divisors, or test whether 141 and 139 share any common factor with 140. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 141. The number 141

This chapter is about the number 141. In Roman numerals it is written CXLI, in binary it is 10001101, and in base eight it is 215. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

141 is a composite number. Its divisors are 1, 3, 47, 141, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 47. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 141 that are smaller than 141 is 51. Because this sum is smaller than 141, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 141 is 19881 and its cube is 2803221. 141 is not a perfect square: it lies between 121 and 144. The sum of its digits is 6, so 141 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 141 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 141: find two whole numbers whose product is 282, then two whole numbers whose sum is 148. Then write 141 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 141 is even. A hint: divisors usually come in pairs whose product is 141, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 141 that has the same number of divisors, or test whether 142 and 140 share any common factor with 141. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 142. The number 142

This chapter is about the number 142. In Roman numerals it is written CXLII, in binary it is 10001110, and in base eight it is 216. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

142 is a composite number. Its divisors are 1, 2, 71, 142, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 71. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 142 that are smaller than 142 is 74. Because this sum is smaller than 142, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 142 is 20164 and its cube is 2863288. 142 is not a perfect square: it lies between 121 and 144. The sum of its digits is 7, so 142 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 142 is even, so it can be shared equally between two people.

Exercise for chapter 142: find two whole numbers whose product is 284, then two whole numbers whose sum is 149. Then write 142 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 142 is even. A hint: divisors usually come in pairs whose product is 142, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 142 that has the same number of divisors, or test whether 143 and 141 share any common factor with 142. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 143. The number 143

This chapter is about the number 143. In Roman numerals it is written CXLIII, in binary it is 10001111, and in base eight it is 217. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

143 is a composite number. Its divisors are 1, 11, 13, 143, so it has 4 divisors in total. As a product of prime numbers it is written 11 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 143 that are smaller than 143 is 25. Because this sum is smaller than 143, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 143 is 20449 and its cube is 2924207. 143 is not a perfect square: it lies between 121 and 144. The sum of its digits is 8, so 143 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 143 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 143: find two whole numbers whose product is 286, then two whole numbers whose sum is 150. Then write 143 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 143 is even. A hint: divisors usually come in pairs whose product is 143, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 143 that has the same number of divisors, or test whether 144 and 142 share any common factor with 143. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 144. The number 144

This chapter is about the number 144. In Roman numerals it is written CXLIV, in binary it is 10010000, and in base eight it is 220. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

144 is a composite number. Its divisors are 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144, so it has 15 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 3 × 3. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 144 that are smaller than 144 is 259. Because this sum is larger than 144, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 144 is 20736 and its cube is 2985984. 144 is itself a perfect square, because 12 × 12 = 144. The sum of its digits is 9, so 144 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 144 is even, so it can be shared equally between two people.

Exercise for chapter 144: find two whole numbers whose product is 288, then two whole numbers whose sum is 151. Then write 144 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 144 is odd. A hint: divisors usually come in pairs whose product is 144, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 144 that has the same number of divisors, or test whether 145 and 143 share any common factor with 144. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 145. The number 145

This chapter is about the number 145. In Roman numerals it is written CXLV, in binary it is 10010001, and in base eight it is 221. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

145 is a composite number. Its divisors are 1, 5, 29, 145, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 29. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 145 that are smaller than 145 is 35. Because this sum is smaller than 145, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 145 is 21025 and its cube is 3048625. 145 is not a perfect square: it lies between 144 and 169. The sum of its digits is 10, so 145 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 145 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 145: find two whole numbers whose product is 290, then two whole numbers whose sum is 152. Then write 145 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 145 is even. A hint: divisors usually come in pairs whose product is 145, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 145 that has the same number of divisors, or test whether 146 and 144 share any common factor with 145. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 146. The number 146

This chapter is about the number 146. In Roman numerals it is written CXLVI, in binary it is 10010010, and in base eight it is 222. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

146 is a composite number. Its divisors are 1, 2, 73, 146, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 73. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 146 that are smaller than 146 is 76. Because this sum is smaller than 146, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 146 is 21316 and its cube is 3112136. 146 is not a perfect square: it lies between 144 and 169. The sum of its digits is 11, so 146 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 146 is even, so it can be shared equally between two people.

Exercise for chapter 146: find two whole numbers whose product is 292, then two whole numbers whose sum is 153. Then write 146 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 146 is even. A hint: divisors usually come in pairs whose product is 146, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 146 that has the same number of divisors, or test whether 147 and 145 share any common factor with 146. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 147. The number 147

This chapter is about the number 147. In Roman numerals it is written CXLVII, in binary it is 10010011, and in base eight it is 223. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

147 is a composite number. Its divisors are 1, 3, 7, 21, 49, 147, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 7 × 7. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 147 that are smaller than 147 is 81. Because this sum is smaller than 147, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 147 is 21609 and its cube is 3176523. 147 is not a perfect square: it lies between 144 and 169. The sum of its digits is 12, so 147 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 147 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 147: find two whole numbers whose product is 294, then two whole numbers whose sum is 154. Then write 147 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 147 is even. A hint: divisors usually come in pairs whose product is 147, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 147 that has the same number of divisors, or test whether 148 and 146 share any common factor with 147. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 148. The number 148

This chapter is about the number 148. In Roman numerals it is written CXLVIII, in binary it is 10010100, and in base eight it is 224. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

148 is a composite number. Its divisors are 1, 2, 4, 37, 74, 148, so it has 6 divisors in total. As a product of prime numbers it is written 2 × 2 × 37. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 148 that are smaller than 148 is 118. Because this sum is smaller than 148, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 148 is 21904 and its cube is 3241792. 148 is not a perfect square: it lies between 144 and 169. The sum of its digits is 13, so 148 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 148 is even, so it can be shared equally between two people.

Exercise for chapter 148: find two whole numbers whose product is 296, then two whole numbers whose sum is 155. Then write 148 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 148 is even. A hint: divisors usually come in pairs whose product is 148, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 148 that has the same number of divisors, or test whether 149 and 147 share any common factor with 148. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 149. The number 149

This chapter is about the number 149. In Roman numerals it is written CXLIX, in binary it is 10010101, and in base eight it is 225. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

149 is a prime number. Its only divisors are 1 and 149. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 149 counters in equal rows, the only way is a single row of 149. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 149 is one of them.

The sum of the divisors of 149 that are smaller than 149 is 1. Because this sum is smaller than 149, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 149 is 22201 and its cube is 3307949. 149 is not a perfect square: it lies between 144 and 169. The sum of its digits is 14, so 149 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 149 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 149: find two whole numbers whose product is 298, then two whole numbers whose sum is 156. Then write 149 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 149 is even. A hint: divisors usually come in pairs whose product is 149, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 149 that has the same number of divisors, or test whether 150 and 148 share any common factor with 149. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 150. The number 150

This chapter is about the number 150. In Roman numerals it is written CL, in binary it is 10010110, and in base eight it is 226. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

150 is a composite number. Its divisors are 1, 2, 3, 5, 6, 10, 15, 25, 30, 50, 75, 150, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 3 × 5 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 150 that are smaller than 150 is 222. Because this sum is larger than 150, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 150 is 22500 and its cube is 3375000. 150 is not a perfect square: it lies between 144 and 169. The sum of its digits is 6, so 150 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 150 is even, so it can be shared equally between two people.

Exercise for chapter 150: find two whole numbers whose product is 300, then two whole numbers whose sum is 157. Then write 150 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 150 is even. A hint: divisors usually come in pairs whose product is 150, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 150 that has the same number of divisors, or test whether 151 and 149 share any common factor with 150. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 151. The number 151

This chapter is about the number 151. In Roman numerals it is written CLI, in binary it is 10010111, and in base eight it is 227. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

151 is a prime number. Its only divisors are 1 and 151. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 151 counters in equal rows, the only way is a single row of 151. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 151 is one of them.

The sum of the divisors of 151 that are smaller than 151 is 1. Because this sum is smaller than 151, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 151 is 22801 and its cube is 3442951. 151 is not a perfect square: it lies between 144 and 169. The sum of its digits is 7, so 151 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 151 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 151: find two whole numbers whose product is 302, then two whole numbers whose sum is 158. Then write 151 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 151 is even. A hint: divisors usually come in pairs whose product is 151, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 151 that has the same number of divisors, or test whether 152 and 150 share any common factor with 151. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 152. The number 152

This chapter is about the number 152. In Roman numerals it is written CLII, in binary it is 10011000, and in base eight it is 230. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

152 is a composite number. Its divisors are 1, 2, 4, 8, 19, 38, 76, 152, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 19. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 152 that are smaller than 152 is 148. Because this sum is smaller than 152, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 152 is 23104 and its cube is 3511808. 152 is not a perfect square: it lies between 144 and 169. The sum of its digits is 8, so 152 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 152 is even, so it can be shared equally between two people.

Exercise for chapter 152: find two whole numbers whose product is 304, then two whole numbers whose sum is 159. Then write 152 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 152 is even. A hint: divisors usually come in pairs whose product is 152, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 152 that has the same number of divisors, or test whether 153 and 151 share any common factor with 152. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 153. The number 153

This chapter is about the number 153. In Roman numerals it is written CLIII, in binary it is 10011001, and in base eight it is 231. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

153 is a composite number. Its divisors are 1, 3, 9, 17, 51, 153, so it has 6 divisors in total. As a product of prime numbers it is written 3 × 3 × 17. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 153 that are smaller than 153 is 81. Because this sum is smaller than 153, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 153 is 23409 and its cube is 3581577. 153 is not a perfect square: it lies between 144 and 169. The sum of its digits is 9, so 153 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 153 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 153: find two whole numbers whose product is 306, then two whole numbers whose sum is 160. Then write 153 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 153 is even. A hint: divisors usually come in pairs whose product is 153, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 153 that has the same number of divisors, or test whether 154 and 152 share any common factor with 153. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 154. The number 154

This chapter is about the number 154. In Roman numerals it is written CLIV, in binary it is 10011010, and in base eight it is 232. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

154 is a composite number. Its divisors are 1, 2, 7, 11, 14, 22, 77, 154, so it has 8 divisors in total. As a product of prime numbers it is written 2 × 7 × 11. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 154 that are smaller than 154 is 134. Because this sum is smaller than 154, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 154 is 23716 and its cube is 3652264. 154 is not a perfect square: it lies between 144 and 169. The sum of its digits is 10, so 154 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 154 is even, so it can be shared equally between two people.

Exercise for chapter 154: find two whole numbers whose product is 308, then two whole numbers whose sum is 161. Then write 154 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 154 is even. A hint: divisors usually come in pairs whose product is 154, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 154 that has the same number of divisors, or test whether 155 and 153 share any common factor with 154. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 155. The number 155

This chapter is about the number 155. In Roman numerals it is written CLV, in binary it is 10011011, and in base eight it is 233. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

155 is a composite number. Its divisors are 1, 5, 31, 155, so it has 4 divisors in total. As a product of prime numbers it is written 5 × 31. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 155 that are smaller than 155 is 37. Because this sum is smaller than 155, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 155 is 24025 and its cube is 3723875. 155 is not a perfect square: it lies between 144 and 169. The sum of its digits is 11, so 155 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 155 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 155: find two whole numbers whose product is 310, then two whole numbers whose sum is 162. Then write 155 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 155 is even. A hint: divisors usually come in pairs whose product is 155, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 155 that has the same number of divisors, or test whether 156 and 154 share any common factor with 155. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 156. The number 156

This chapter is about the number 156. In Roman numerals it is written CLVI, in binary it is 10011100, and in base eight it is 234. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

156 is a composite number. Its divisors are 1, 2, 3, 4, 6, 12, 13, 26, 39, 52, 78, 156, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 3 × 13. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 156 that are smaller than 156 is 236. Because this sum is larger than 156, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 156 is 24336 and its cube is 3796416. 156 is not a perfect square: it lies between 144 and 169. The sum of its digits is 12, so 156 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 156 is even, so it can be shared equally between two people.

Exercise for chapter 156: find two whole numbers whose product is 312, then two whole numbers whose sum is 163. Then write 156 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 156 is even. A hint: divisors usually come in pairs whose product is 156, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 156 that has the same number of divisors, or test whether 157 and 155 share any common factor with 156. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 157. The number 157

This chapter is about the number 157. In Roman numerals it is written CLVII, in binary it is 10011101, and in base eight it is 235. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

157 is a prime number. Its only divisors are 1 and 157. A prime number cannot be arranged as a rectangle of dots with more than one row and more than one column: if you try to place 157 counters in equal rows, the only way is a single row of 157. Mathematicians have known since ancient Greece that there are infinitely many prime numbers, and 157 is one of them.

The sum of the divisors of 157 that are smaller than 157 is 1. Because this sum is smaller than 157, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 157 is 24649 and its cube is 3869893. 157 is not a perfect square: it lies between 144 and 169. The sum of its digits is 13, so 157 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 157 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 157: find two whole numbers whose product is 314, then two whole numbers whose sum is 164. Then write 157 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 157 is even. A hint: divisors usually come in pairs whose product is 157, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 157 that has the same number of divisors, or test whether 158 and 156 share any common factor with 157. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 158. The number 158

This chapter is about the number 158. In Roman numerals it is written CLVIII, in binary it is 10011110, and in base eight it is 236. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

158 is a composite number. Its divisors are 1, 2, 79, 158, so it has 4 divisors in total. As a product of prime numbers it is written 2 × 79. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 158 that are smaller than 158 is 82. Because this sum is smaller than 158, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 158 is 24964 and its cube is 3944312. 158 is not a perfect square: it lies between 144 and 169. The sum of its digits is 14, so 158 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 158 is even, so it can be shared equally between two people.

Exercise for chapter 158: find two whole numbers whose product is 316, then two whole numbers whose sum is 165. Then write 158 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 158 is even. A hint: divisors usually come in pairs whose product is 158, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 158 that has the same number of divisors, or test whether 159 and 157 share any common factor with 158. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 159. The number 159

This chapter is about the number 159. In Roman numerals it is written CLIX, in binary it is 10011111, and in base eight it is 237. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

159 is a composite number. Its divisors are 1, 3, 53, 159, so it has 4 divisors in total. As a product of prime numbers it is written 3 × 53. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 159 that are smaller than 159 is 57. Because this sum is smaller than 159, the number is called deficient. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 159 is 25281 and its cube is 4019679. 159 is not a perfect square: it lies between 144 and 169. The sum of its digits is 15, so 159 is divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 159 is odd, so sharing it equally between two people always leaves one left over.

Exercise for chapter 159: find two whole numbers whose product is 318, then two whole numbers whose sum is 166. Then write 159 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 159 is even. A hint: divisors usually come in pairs whose product is 159, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 159 that has the same number of divisors, or test whether 160 and 158 share any common factor with 159. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.

Chapter 160. The number 160

This chapter is about the number 160. In Roman numerals it is written CLX, in binary it is 10100000, and in base eight it is 240. Writing the same number in different bases does not change the number itself; it only changes the symbols we use to record it. A base tells us how many different digits we use before we start a new column.

160 is a composite number. Its divisors are 1, 2, 4, 5, 8, 10, 16, 20, 32, 40, 80, 160, so it has 12 divisors in total. As a product of prime numbers it is written 2 × 2 × 2 × 2 × 2 × 5. You can check this by multiplying the factors back together. This way of breaking a number into primes is unique, which is why primes are often called the building blocks of the whole numbers.

The sum of the divisors of 160 that are smaller than 160 is 218. Because this sum is larger than 160, the number is called abundant. Perfect numbers are rare: among the numbers in this handbook only 6 and 28 are perfect. Checking a claim like this by hand is a good habit: add up the smaller divisors and compare the total with the number.

The square of 160 is 25600 and its cube is 4096000. 160 is not a perfect square: it lies between 144 and 169. The sum of its digits is 7, so 160 is not divisible by 3; this is the divisibility rule for 3, which works because 10 leaves a remainder of 1 when divided by 3. 160 is even, so it can be shared equally between two people.

Exercise for chapter 160: find two whole numbers whose product is 320, then two whole numbers whose sum is 167. Then write 160 as a sum of powers of two, using the binary form above: each 1 in the binary form stands for one power of two. Finally, explain in your own words why the number of divisors of 160 is even. A hint: divisors usually come in pairs whose product is 160, and a pair can only collapse into a single divisor when the number is a perfect square.

Notes for the teacher: the ideas in this chapter connect to place value, multiplication tables and the idea of proof. Students who finish early can look for the next number after 160 that has the same number of divisors, or test whether 161 and 159 share any common factor with 160. Two numbers next to each other never share a factor larger than 1, which is a short and beautiful argument students can discover themselves.